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Agriculture And Veterinary
Medicine With International
Participation
Proceedings book***



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November 18, 2021*



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POLJOPRIVREDNI FAKULTET**

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SA 45. SMOTRE NAUČNIH RADOVA STUDENATA POLJOPRIVREDE I
VETERINARSKE MEDICINE SA MEĐUNARODNIM UČEŠĆEM**

**PROCEEDINGS
OF THE 45th CONFERENCE FOR STUDENTS OF AGRICULTURE AND
VETERINARY MEDICINE
WITH INTERNATIONAL PARTICIPATION**

18 November 2021

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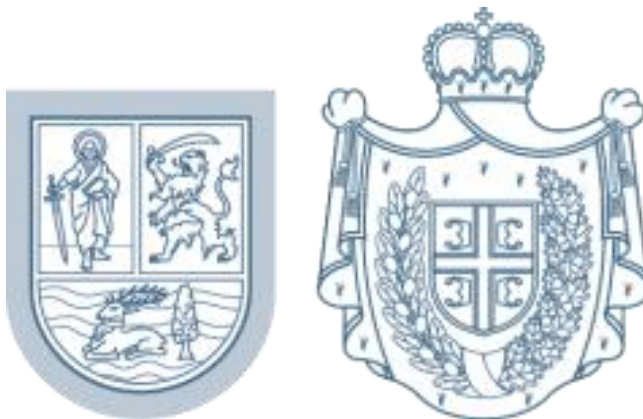
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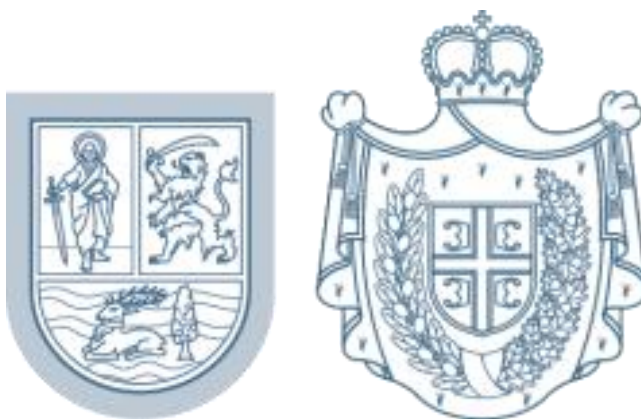


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**45th CONFERENCE FOR STUDENTS OF AGRICULTURE AND VETERINARY
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Council of the University of Novi Sad, Faculty of Agriculture, at its fourth meeting on 18th of February 1981 made a decision that the Conference for students of agriculture and veterinary medicine with international participation, held each year at the Faculty of Agriculture will be held in the memory of academician ***Dr Petar Drezgić***, professor, and the prizes awarded at the Conference will bear his name.

PREFACE

The first Conference for students of agriculture and veterinary medicine with international participation was held in Novi Sad 45 years ago. To the present date, many papers from national and international authors were presented (Hungary – Gödölö and Debrecen, Germany – Kassel, Slovakia – Nitra, Poland – Wroclaw and Warsaw, Romania – Timisoara, Cluj and Bucharest, Bulgaria – Plovdiv, Russia – Moscow and Kemerovo, Macedonia – Skopje and Bitola, Bosnia and Herzegovina – Sarajevo, Banja Luka, Croatia – Zagreb). Since 2009, papers presented at the Conference are published in a special publication named “PROCEEDINGS OF THE 45th CONFERENCE FOR STUDENTS OF AGRICULTURE AND VETERINARY MEDICINE WITH INTERNATIONAL PARTICIPATION” where all published papers have all elements of the original scientific paper. Starting this year, we enabled students and mentors who are interested in benefitting more from this Conference to publish their papers in the journal Contemporary Agriculture published by the University of Novi Sad, Faculty of Agriculture.

Also, besides the Faculty of Agriculture from Novi Sad, co-organizer of the Conference is also Faculty of Agriculture from Timisoara and there is a prospect of joining another University center from the region. This Conference is strongly supported by the Student Parliament of the Faculty of Agriculture, Novi Sad.

As the host and the principal organizer of the Conference University of Novi Sad, Faculty of Agriculture has a special honor and a pleasure to wish a warm welcome to all participants, successful work and amusing socializing in Novi Sad.

Vice-dean for science and international cooperation
Dr Branko Ćupina, full professor

President of the organizing committee
Dr Dragana Budakov, associate professor

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EVALUATION OF RELEVANT MINERAL CONTENT OF MEDICINAL WILD-GROWN MUSHROOM *GANODERMA LUCIDUM* FROM DIFFERENT ROMANIAN AREAS

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Summary: *Edible and medicinal wild-growing mushrooms from around the world are known since ancient times being consumed for their nutritional and flavor properties. Although Ganoderma lucidum is well known for its health beneficial properties including immunomodulatory effects, it naturally accumulates, in addition to essential chemical elements, some potentially toxic or radioactive ones, which could have impact on the human health. The present study aimed to evaluate the mineral content, both of essential chemical elements and of heavy metals and radioactive ones. The study involved collecting samples from three different areas in Transylvania, Romania. In the studied areas, six biominerals were found at relevant concentration, as follow: Si 41.686 mg/kg, Ca 59.260 mg/kg, P 29.756 mg/kg, Mn 36.550 mg/kg, K 171.519 mg/kg and Fe 203.034 mg/kg. The results showed the presence of some heavy metals and radioactive elements. The average concentration of some heavy metals detected in Ganoderma lucidum was of 7.5975 mg/kg for Pb and of 11.689 mg/kg for As from Sibiu area. Samples from Hunedoara showed the presence of Fr 3.324 mg/kg, At 28.960 mg/kg and Po 2.757 mg/kg, while those from Sibiu the presence of U 11.009 mg/kg. The study showed the significant biomineral content of Ganoderma lucidum, and the importance of the evaluation of some potential toxic chemical elements.*

Keywords: *heavy metals, radioactive elements, medicinal mushroom, Ganoderma lucidum, Romania.*

INTRODUCTION

Romania has a great potential for harvesting edible mushrooms, mainly macrofungi (Vasile *et al.* 2017). Medicinal, wild mushrooms are known for their health benefits due to their content in nutrients and biologically active compounds that help maintain physical and mental health (Wang *et al.* 2014; Chang and Miles, 2004). *Ganoderma lucidum* is one of the most studied mushroom species globally based on its pharmacological properties, such as antiviral, immunostimulatory, anticancer, antioxidant, anti-inflammatory and antimicrobial (Zeng *et al.* 2012; Savin *et al.* 2020). *Ganoderma lucidum* grows in an optimal habitat of oak and beech forests, conditions widely found in Romania (Copot and Tanase 2017), being collected and used in traditional medicine. The chemical composition of mushrooms varies from one species to another and depends on the substrate on which they grow (Kalač, 2009, 2010; Bernaś *et al.* 2006). *Ganoderma lucidum* is considered an important source of nitrogen, phosphorus, potassium, calcium, magnesium and trace elements (Peksen *et al.* 2011). Biomineral composition of mushrooms vary according to the species (Wani *et al.* 2010). In a study on *Ganoderma sp.* collected from Chaukori, Pithoragarh (Singh *et al.* 2020), the concentrations of biomineral elements identified with X-ray fluorescence spectrometry were found as follow: Al 3.79 mg/kg, Ca 5400 mg/kg, Cl 399 mg/kg, Cr 2.01 mg/kg, Cu 26.2 mg/kg, Fe 550 mg/kg, K 5601mg/kg, Mg 1301 mg/kg, Mn 48.3 mg/kg, Mo 7.95 mg/kg, Na 500 mg/kg, Ni 7.23 mg/kg, P 2101 mg/kg, Rb 5.57 mg/kg, Ru 5.10 mg/kg, S 1699 mg/kg, Si 460 mg/kg, Sr 6.02 mg/kg, Ti 20.0 mg/kg, Zn 27.2 mg/kg, Zr 4.97 mg/kg. Instead, mushrooms have the ability to store high concentrations of heavy metals and radioactive elements, as well, acting as efficient bioindicators (Giovani *et al.* 1990; Ingrao *et al.* 1992; Rühm *et al.* 1997). In recent years, several studies have been conducted in Romania with regard to the content of heavy metals in the fruiting body of fungi, significant accumulations of Fe, Cu, Zn and Pb being detected (Stihi *et al.* 2011; Radulescu *et al.* 2010; Zsigmond *et al.* 2015). Regarding the radioactive elements which may potentially contaminate mushrooms, the first studies were related to the effects of the Chernobyl explosion on the edible species *Boletus edulis*, *Cantharellus cibarius*, *Xerocomus badius*, (Elstner *et al.* 1989; Elstner *et al.* 1987; Battiston *et al.* 1989). The aim of the present study was to determine the mineral content of samples of wild *Ganoderma lucidum* collected from three different Romanian areas, both of essential/non-essential elements and toxic ones (heavy metals, radioactive). SEM analysis was also performed for identification of the species.

MATERIALS AND METHODS

MUSHROOM COLLECTION

The material used to carry out this study was the fruiting body of the mushrooms from *Ganoderma lucidum* species collected in the fall of 2016 from three different areas of Transylvania, Romania, as follow: AB - Alba/Petrești Forest, HD - Hunedoara/Brănișca Forest and SB - Sibiu/Dumbrava Forest. Areas were chosen according to the fact that they are

most commonly used by people for collecting this type of mushroom for personal use. Samples were collected in containers and marked with specific GPS coordinates. Number of samples which was collected is Alba (2), Hunedoara (2) and Sibiu (4). This small number of samples is due to the fact that this species is very rare. The study areas are located from north to east as follows: Sibiu (N 45.75947 - E 24.12532); Alba (N 45.9212 - E 23.52293); Hunedoara (N 45.93538 - E 22.74500). Samples were collected from mixed forests beech and oak. The wild mushrooms were collected randomized in the mature stage, the second phase of secondary metabolism, respectively, at which mushrooms accumulate useful chemical compounds. Wild *Ganoderma lucidum* have been identified by using the specific and confirmed by SEM investigation.

SEM (SCANNING ELECTRON MICROSCOPY) ANALYSIS

Before SEM analysis, water was removed from the tissues by dehydration (twice ethanol washings with 25% ETOH/75% H₂O and 50% ETOH/50% H₂O, respectively). Samples were left to dry at room temperature, protected from possible contamination with dust particles from atmosphere. Because biological samples show poor electrical conductivity, in order to prevent the phenomena of electric charge and their distortion by the electron beam, mushrooms were treated with a thin layer of gold. Morphological characterization of treated *Ganoderma lucidum* samples was evaluated by SEM analysis using a Jeol JSM 66-10LV Scanning Electron Microscopy.

DETERMINATION OF THE MINERAL CONTENT BY XRF

Elemental analysis was performed by X-ray fluorescence spectrometry (XRF) using the Supermini XRF (Rigaku) spectrometer and the method described by Costa *et al.* 2019. The values of mineral oxides provided by the obtained spectra were calculated based on ash content of mushroom samples, previously dried at room temperature for a few days and ground using a non-metallic mill.

STATISTICAL ANALYSIS

Statistical analysis of multivariate data was performed using Canoco 5.12 software for the principal component analysis (PCA) (Lepš and Šmilauer 2003) regarding the chemical elements identified in mushroom samples.

RESULTS AND DISCUSSION

MORPHOLOGICAL CHARACTERISTICS OF *GANODERMA LUCIDUM*

Using the SEM analysis, mycelium and spores of *Ganoderma lucidum* were examined, in order to identify the species and to confirm the maturity stage of collected mushrooms. Morphological characteristics are well highlighted, such as the hymenial layer in the form of

microtubes and ellipsoidal spores specific to the species. Fruit body have dimension of 10-30 cm with kidney shaped, stem 5-20 cm excentric. Pore surface is ochraceous, rounded, with 4-6 mm, tubes 30 mm deep, spores ellipsoidal, flattened at one end with dimension 7-12 x 6-8 μm (Stefan, 2012). The presence of the spores in the hymenial layer of *Ganoderma lucidum* sample, as shown in Figures 1-2, confirmed the stage of maturity of the fruiting body, as it begins to eliminate spores.

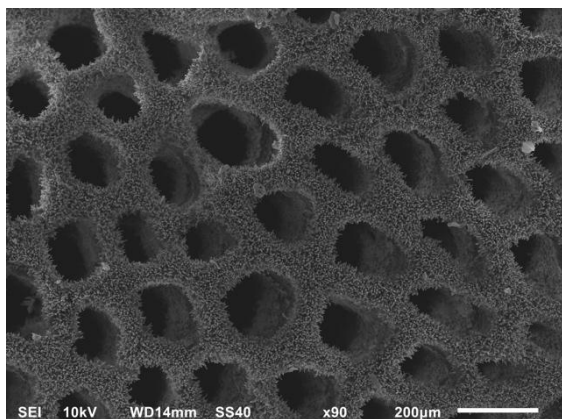


Figure 1. SEM image of *Ganoderma lucidum* microtubes (x 90)

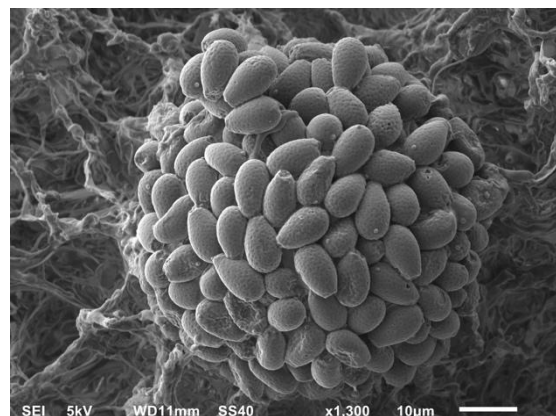


Figure 2. SEM image of *Ganoderma lucidum* spores (x 1300)

MINERAL CONTENT OF *GANODERMA LUCIDUM* FROM DIFFERENT GEOGRAPHICAL ROMANIAN AREAS

Considering the significance of the determination of essential, non-essential and potential toxic elements in mushrooms intended for human consumption, *Ganoderma lucidum* samples collected from three different Romanian areas (AB - Alba, HD - Hunedoara, SB - Sibiu) were investigated for their mineral content.

Figure 3 indicates the content of essential macroelements (Ca, P), oligoelements (K, S), microelements (Fe, Mn, Si, Ni) and non-essential chemical elements (Ag, Sr). As expected, potassium had the highest concentration among the identified elements, followed by iron, silver, calcium and manganese.

A number of ten elements were commonly found in samples from all investigated areas, Si, P, Fe, Ag, Ca, K, Mn, Sr, S, and Ni. The most significant mineral content of *Ganoderma lucidum* from Sibiu area was found for Fe (203.034 mg/kg), Ag (173.291 mg/kg), Zn (26.637 mg/kg), from Alba area, K (171.519 mg/kg), P (29.756 mg/kg), Cu (281.685 mg/kg), and from Hunedoara area, mainly for Ti (51.149 mg/kg). In a study of Singh *et al.* (2020) on *Ganoderma lucidum* collected from Chaukori, Pithoragarh, the concentration of Fe was 550 mg/kg, Zn 27.2 mg/kg, K 5601 mg/kg, P 2101 mg/kg, Cu 26.2 mg/kg, Ti 20.0 mg/kg, while Ag was not detected.

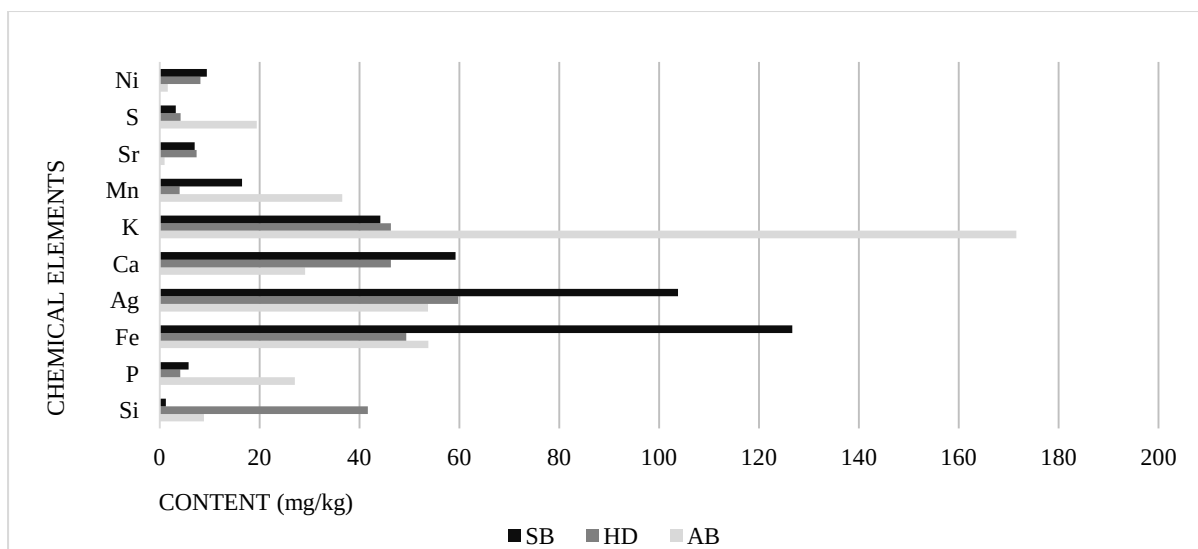


Figure 3. Average mineral content of identified chemical elements in *Ganoderma lucidum* from studied areas

In the present study, a number of 40 chemical elements Ag, Al, As, At, Br, Ca, Cl, Co, Cu, Eu, Fe, Fr, Ga, Gd, Ge, Ir, K, Lu, Mn, Mo, Mg, Ni, Os, P, Pb, Po, Pt, Rb, Re, S, Se, Si, Sr, Ti, Tm, U, V, W, Y, Zn was detected in the selected samples.

The concentration of other elements identified in *Ganoderma lucidum* from studied area, are shown in Table 1. Among the studied areas, the samples from the forest of Alba County had the highest potassium content, followed by those from Hunedoara and Sibiu. According to the obtained experimental data, strontium content of samples from Alba County (0.981 mg/kg) and sulphur from Sibiu County (0.698 mg/kg) was the lowest. Iron was founded in high concentration in samples from in Sibiu County (203.034 mg/kg), followed by silver (173.291 mg/kg) and calcium (59.260 mg/kg).

In the present study, As and Pb were not found in the samples collected from Alba and Hunedoara, while Pb was detected in concentration of 7.597 mg/kg in samples from Sibiu. Hunedoara is representative for the accumulation of the radioactive elements, Fr (3.324 mg/kg), At (28.960 mg/kg) and Po (2.757 mg/kg), followed by Sibiu with U levels of 11.009 mg/kg. These radioactive elements can be explained by the fact that in this area, a military training area was set. Also, there is a potential risk associated to the accumulation of heavy metals and radioactive elements from food or pharmaceutical chain (Wang *et al.* 2017).

Accumulation of several heavy metals in different mushrooms from Romanian areas has been shown by other studies. Higher lead content (3.47mg/kg) was reported in *Lycoperdon perlatum* samples from Dâmbovița County (Stihi *et al.* 2011). Mushrooms can be used as bioindicators in studies of monitoring the soil pollution with heavy metals or radioactive elements (Battiston *et al.* 1989). The admitted levels of lead per day according to World Health Organisation (WHO) are 0.30 mg/kg/day, while for arsenic of 0.3 µg/kg/day. Another study on concentration of heavy metals in *Ganoderma lucidum* cultivated on different substrate compositions, showed the Pb content of 1.2 mg/kg in the fruiting bodies in this mushroom (Siwulski *et al.* 2019). In another study from Pakistan, lead was not detected in wild *Ganoderma lucidum* (Sharif *et al.* 2016). Similarly, a study by Singh *et al.* (2020) on

Ganoderma lucidum collected from forest of India, Uttarakhand, did not report the presence of lead, but this heavy metal was found in other species of *Ganoderma*, *Ganoderma brownie* with Pb concentration of 3.93 mg/kg and *Ganoderma philippii* with Pb concentration of 4.67 mg/kg.

Table 1. Content of other chemical elements (mg/kg) identified in *Ganoderma lucidum* from studied area

Area	AB1	AB2	HD1	HD2	SB1	SB2	SB3	SB4
Al	n.d.	3.643	n.d.	12.787	n.d.	n.d.	n.d.	n.d.
As	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	11.689
At	n.d.	n.d.	28.960	n.d.	n.d.	n.d.	n.d.	n.d.
Br	n.d.	n.d.	n.d.	n.d.	4.317	n.d.	n.d.	n.d.
Cl	n.d.	1.945	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
Co	n.d.	0.962	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
Cu	281.685	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
Eu	n.d.	n.d.	n.d.	8.663	n.d.	n.d.	n.d.	n.d.
Fr	n.d.	n.d.	n.d.	3.324	n.d.	n.d.	n.d.	n.d.
Ga	n.d.	1.564	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
Gd	n.d.	7.964	n.d.	n.d.	n.d.	26.519	n.d.	n.d.
Ge	n.d.	n.d.	n.d.	1.206	n.d.	4.892	n.d.	18.991
Ir	n.d.	n.d.	n.d.	3.098	n.d.	n.d.	n.d.	n.d.
Lu	n.d.	n.d.	n.d.	4.331	n.d.	n.d.	n.d.	n.d.
Mg	n.d.	5.182	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
Mo	n.d.	n.d.	n.d.	76.143	n.d.	n.d.	n.d.	n.d.
Os	n.d.	n.d.	n.d.	5.694	n.d.	n.d.	n.d.	n.d.
Pb	n.d.	n.d.	n.d.	n.d.	7.597	n.d.	n.d.	n.d.
Po	n.d.	n.d.	n.d.	2.757	n.d.	n.d.	n.d.	n.d.
Pt	n.d.	n.d.	n.d.	2.041	n.d.	n.d.	n.d.	n.d.
Rb	n.d.	1.138	n.d.	n.d.	n.d.	n.d.	n.d.	16.263
Re	n.d.	n.d.	48.204	n.d.	n.d.	n.d.	n.d.	n.d.
Se	n.d.	n.d.	n.d.	n.d.	4.796	n.d.	n.d.	n.d.
Ti	n.d.	n.d.	51.149	n.d.	25.559	18.439	n.d.	n.d.
Tm	n.d.	2.093	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.
U	n.d.	n.d.	n.d.	n.d.	11.009	n.d.	n.d.	n.d.
V	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	47.754	n.d.
W	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	26.304	n.d.
Y	n.d.	n.d.	n.d.	n.d.	n.d.	n.d.	9.591	n.d.
Zn	23.663	n.d.	n.d.	n.d.	6.510	6.711	12.816	26.637

n.d. – not detected

Figure 4 shows a PCA plot of the multivariate data of mineral content of mushroom samples from different Romanian areas. No significant correlations were found between zone

altitude and chemical elements, between geographical coordinates and chemical elements, showing that spatial coordinates and altitude are not significant predictors of the mineral content. The results of the PCA analysis showed a grouping of samples indicating the variability of the chemical composition depending on the plots and area. Sample 1 corresponds with Alba area, and 7, 8 corresponds Sibiu area. This area is grouped showing similarity in the chemical composition, as well as sample 3 Hunedoara area and 5 Sibiu area, 2 Alba area and 6 Sibiu area. The only isolated sample that does not show similar values was the 4 Hunedoara area, indicating the presence of different chemical elements, and the largest value of aluminium. A positive correlation of Zn, Fe, Mn, As, U and Pb have been found for samples of Sibiu area. Similarly, correlations of Os, Eu, Po, Fr, Ir, Pt, Lu, Eu and Mo of samples from 4 area, Hunedoara, were found, linked to the previously set military area in that region.

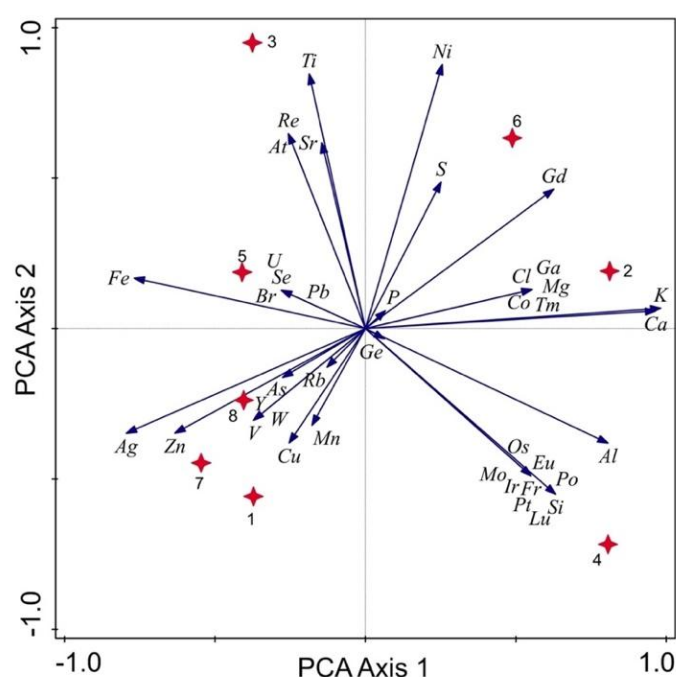


Figure 4. PCA plot of chemical elements identified in *Ganoderma lucidum* mushroom from Romanian areas

The study of Fogarasi *et al.* (2020) on five mushrooms species (*Agaricus bisporus*, *Pleurotus ostreatus*, *Cantharellus cibarius*, *Boletus edulis*, *Lactarius piperatus*) collected in 2018 from Transylvania region, Meseş Mountain, Sălaj Country, showed the presence of 11 chemical elements (Cu, Fe, Mn, Cr, Ba, Co, Zn, Ca, Mg, Na, K) detected. Their results revealed that *Cantharellus cibarius* had the highest concentration of Cu (49.20 mg/kg), Fe (165.50 mg/kg) and Ba (1.00 mg/kg), follow by *Boletus edulis* with Mn (60.84 mg/kg) and Co (0.32 mg/kg), and *Agaricus bisporus* which has a significant content of Cr (0.85 mg/kg). The mentioned study did not report the presence of heavy metals or radioactive elements in the investigated samples.

CONCLUSIONS

This paper evaluated the elemental composition of wild medicinal *Ganoderma lucidum* from several Transylvanian forests. This first investigation confirmed the presence of valuable chemical elements in *Ganoderma lucidum* samples collected from three different Romanian geographical regions. Using the SEM analysis, the stage of maturity of the fruiting body was confirmed. Some potentially toxic and radioactive elements were identified in the fruiting body of the fungus. The analysis of the mineral content of *Ganoderma lucidum* investigated samples showed the presence of 40 chemical elements and 10 common chemical elements. This mushroom can serve as a dietary source of calcium, iron, manganese, zinc, silicon and potassium, and therefore, makes a remarkable contribution in the nutritional diet for humans. But, on the other side, these mushrooms may accumulate toxic elements such as Pb and As as hereby found, so that evaluation of potential toxic metals should be conducted.

REFERENCES

BATTISTON, G. A., DEGETTO, S., GERBANSI, R., & SBRIGNADELLO, G. (1989). Radioactivity in mushrooms in northeast Italy following the Chernobyl accident. *Journal of Environmental Radioactivity*, 9(1), 53-60.

BERNAŚ, E., JAWORSKA, G., & LISIEWSKA, Z. (2006). Edible mushrooms as a source of valuable nutritive constituents. *Acta Scientiarum Polonorum Technologia Alimentaria*, 5(1), 5-20.

COPOT, O., & TANASE, C. (2017). Maxent modelling of the potential distribution of *Ganoderma lucidum* in North-Eastern Region of Romania. *Journal of Plant Development*, 24.

COSTA, V. C., AMORIM, F. A. C., DE BABOS, D. V., & PEREIRA-FILHO, E. R. (2019). Direct determination of Ca, K, Mg, Na, P, S, Fe and Zn in bivalve mollusks by wavelength dispersive X-ray fluorescence (WDXRF) and laser-induced breakdown spectroscopy (LIBS). *Food chemistry*, 273, 91-98.

ELSTNER, E. F., FINK, R., HÖLL, W., LENGFELDER, E., & ZIEGLER, H. (1989). Radioactivity in mushrooms, mosses and soil samples of defined biotops in SW Bavaria-two years after "Chernobyl". *Oecologia*, 80(2), 173-177.

ELSTNER, E. F., FINK, R., HÖLL, W., LENGFELDER, E., & ZIEGLER, H. (1987). Natural and Chernobyl-caused radioactivity in mushrooms, mosses and soil-samples of defined biotops in SW Bavaria. *Oecologia*, 73(4), 553-558.

GIOVANI, C., PADOVANI, R., & NIMIS, P. L. (1990). Investigation of the performance of macromycetes as bioindicators of radioactive contamination. In *Transfer of radionuclides in natural and semi-natural environments*.

INGRAO, G., BELLONI, P., & SANTARONI, G. P. (1992). Mushrooms as biological monitors of trace elements in the environment. *Journal of radioanalytical and nuclear chemistry*, 161(1), 113-120.

KALAČ, P. (2009). Chemical composition and nutritional value of European species of wild growing mushrooms: A review. *Food chemistry*, 113(1), 9-16.

KALAČ, P. (2010). Trace element contents in European species of wild growing edible mushrooms: a review for the period 2000–2009. *Food chemistry*, 122(1), 2-15.

LEPŠ, J., & ŠMILAUER, P. (2003). *Multivariate analysis of ecological data using CANOCO*. Cambridge university press.

MILES, P. G., & CHANG, S. T. (2004). *Mushrooms: cultivation, nutritional value, medicinal effect, and environmental impact*. CRC press.

PEKSEN, A., YAKUPOGLU, G., YAKUPOGLU, T., GULSER, C., OZTURK, E., & OZDEMIR, N. (2011). Changes in chemical compositions of substrates before and after *Ganoderma lucidum* cultivation. *World Journal of Microbiology and Biotechnology*, 27(3), 637-642.

RADULESCU, C., STIHI, C., POPESCU, I. V., BUSUIOC, G., GHEBOIANU, A. I., CIMPOCA, V. G., ... & DIACONESCU, M. (2010). Determination of heavy metals content in wild mushrooms and soil by EDXRF and FAAS techniques. *Ovidus Uni. Ann. Chem*, 21(1), 9-14.

RÜHM, W., KAMMERER, L., HIERSCHE, L., & WIRTH, E. (1997). The ¹³⁷Cs/¹³⁴Cs ratio in fungi as an indicator of the major mycelium location in forest soil. *Journal of Environmental Radioactivity*, 35(2), 129-148.

SAVIN, S., CRACIUNESCU, O., OANCEA, A., ILIE, D., CIUCAN, T., ANTOHI, L. S., ... & OANCEA, F. (2020). Antioxidant, Cytotoxic and Antimicrobial Activity of Chitosan Preparations Extracted from *Ganoderma Lucidum* Mushroom. *Chemistry & biodiversity*, 17(7), e2000175.

SHARIF, S., MUSTAFA, G., MUNIR, H., WEAVER, C. M., JAMIL, Y., & SHAHID, M. (2016). Proximate composition and micronutrient mineral profile of wild *Ganoderma lucidum* and four commercial exotic mushrooms by ICP-OES and LIBS. *Journal of Food and Nutrition Research*, 4(11), 703-708.

SINGH, R., KAUR, N., SHRI, R., SINGH, A. P., & DHINGRA, G. S. (2020). Proximate composition and element contents of selected species of *Ganoderma* with reference to dietary intakes. *Environmental monitoring and assessment*, 192(5), 1-15.

SIWULSKI, M., RZYMSKI, P., BUDKA, A., KALAČ, P., BUDZYŃSKA, S., DAWIDOWICZ, L., ... & NIEDZIELSKI, P. (2019). The effect of different substrates on the growth of six cultivated mushroom species and composition of macro and trace elements in their fruiting bodies. *European Food Research and Technology*, 245(2), 419-431.

STEFAN BUCZACKI (2012). *Fungi Guide, The most complete field guide to the mushrooms & toadstools of Britain & Ireland*, Harper Collins Publishers Ltd.

STIHI, C., RADULESCU, C., BUSUIOC, G., POPESCU, I. V., GHEBOIANU, A., & ENE, A. (2011). Studies on accumulation of heavy metals from substrate to edible wild mushrooms. *Romanian Journal of Physics*, 56(1-2), 257-264.

VASILE, D., DINCĂ, L., & ENESCU, C. M. (2017). Impact of collecting mushrooms from the spontaneous flora on forest ecosystems in Romania. *AgroLife Scientific Journal*, 6(1), 268-275.

WANG, X. M., ZHANG, J., WU, L. H., ZHAO, Y. L., LI, T., LI, J. Q., ... & LIU, H. G. (2014). A mini-review of chemical composition and nutritional value of edible wild-grown mushroom from China. *Food chemistry*, 151, 279-285.

WANG, X., LIU, H., ZHANG, J., LI, T., & WANG, Y. (2017). Evaluation of heavy metal concentrations of edible wild-grown mushrooms from China. *Journal of Environmental Science and Health, Part B*, 52(3), 178-183.

WANI, B. A., BODHA, R. H., & WANI, A. H. (2010). Nutritional and medicinal importance of mushrooms. *Journal of Medicinal Plants Research*, 4(24), 2598-2604.

ZENG, X., SUWANDI, J., FULLER, J., DORONILA, A., & NG, K. (2012). Antioxidant capacity and mineral contents of edible wild Australian mushrooms. *Food science and technology international*, 18(4), 367-379.

ZSIGMOND, A. R., VARGA, K., HARANGI, S., BARANYAI, E., & URÁK, I. (2015). Elemental profile of edible mushrooms from a forest near a major Romanian city. *Acta Universitatis Sapientiae, Agriculture and Environment*, 7(1), 98-107.

CASE STUDY: RAINGARDENS AS A SOLUTION TO STORMWATER MANAGEMENT IN NOVI SAD, SERBIA

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Summary: *The problem of floods in urban areas is one of the major ecological problems that climate change brings. Increase of impervious land due to urbanization causes problems of urban flooding in cities like Novi Sad. The goal of the paper is to offer a solution to stormwater management in Novi Sad by proposing a revitalization of a high-risk flood area with the use of raingardens. This model could be applied to other high-risk flood areas in Novi Sad because the use of raingardens in stormwater management is a step to build a sustainable urban area and mitigate the consequences of climate change.*

Key words: *raingardens, stormwater management, urban flooding, Novi Sad, Low Impact Development*

INTRODUCTION

Due to urbanization, increase in impervious land area leads to the increase of floods and runoff water that cannot infiltrate into the soil in urban areas (Wang et al., 2001; Đuraković and Sekulić, 2017). During precipitation of higher intensity, the City of Novi Sad is dealing with urban flooding and pollution because the water outflows from storm drains over the road surface (Greksa et al, 2021). One of the best solutions to the problem of urban flooding is stormwater management combined with urban design (Sharma and Malaviya, 2021). Low Impact Development (LID) is a sustainable approach in stormwater management represented mainly in Northern European countries, the USA, Canada and Australia, which mitigates the negative effects of urbanization and climate change and enhances biodiversity in urban areas (Dietz, 2007; Ishimatsu et al., 2017). A type of LID are raingardens, shallow planted depressions that eliminate run-off through promoting infiltration and pollutant absorption (Dunnet and Clayden, 2007).

The goal of this research is to revitalize and enhance a high-risk flood area in Novi Sad by implementing a raingarden on the green strip beside the road. This model of revitalization could be used on other locations in Novi Sad or Serbia that have a similar problem of flooding.

MATERIAL AND METHODS

Methodology included all aspects of systematization of available data from written literature on rain gardens, which considered insight into suitable resistant plant species for usage in our region and appropriate methods of plant layout. According to that, the research has been focused on the species and varieties suitable for the rain garden, analyzed by their shape, resistance to abiotic and biotic factors and useful value. The history of rain gardens, the impact on the environment and the issues related to rain gardens were also examined factors that are of a great importance in the selection of resistant species. Due to the necessary data on climate and land factors, collecting terrain data on the existing condition at the location for planning was another of the applied methods. The project also contained a cartographic method and method of interviewing experts who have experience in the selection of planting material and approaches to design specific green areas, whose experience is based on years of research. At the same time, the field work was carried out by the authors (Figure 1.). Field data were processed in Adobe Photoshop and Autodesk AutoCAD.



Figure 1. Location of the analyzed area, Partizanska Street, Novi Sad, Serbia

(Source: <https://www.google.rs/maps>, 2021)

RESULTS AND DISCUSSION

Rain Gardens and Bioswales are examples of green infrastructure, also called Low Impact Design, and use natural functions to capture and filter stormwater. Rain gardens and bioswales tackle the issue of pollution in waterways due to urban runoff by absorbing 30% more stormwater into the ground than a conventional lawn and reduce 70% of surface water pollution that originates from stormwater runoff. Sustainable stormwater management facilities such as bioswales and rain gardens are one way in which cities are simultaneously addressing the need to replace or repair stormwater infrastructure while also meeting regulatory obligations. Retrofitting patterns of neighborhood development through the implementation of infrastructure like bioswales is localized solution to stormwater management. Such infrastructure addresses sustainability and resilience goals while reflecting the city as part of rather than separate from the ecosystem. A number of issues point to the

importance of studying how urban residents perceive and interact with sustainable stormwater systems: continued urbanization, “sustainability planning” that focuses upon increased housing densities and compact development (Newman and Jennings, 2008), documented benefits of nature for urban residents (Jordan III, 2000), trends toward green infrastructure solutions (EPA, 2007; Echols and Pennypacker, 2008), and potential social benefits of sustainable stormwater infrastructure as “civic nature” or in contributing to public understanding of the stormwater system (Hough, 2004; Wolf, 2007; Newman and Jennings, 2008) (Figure 2.).

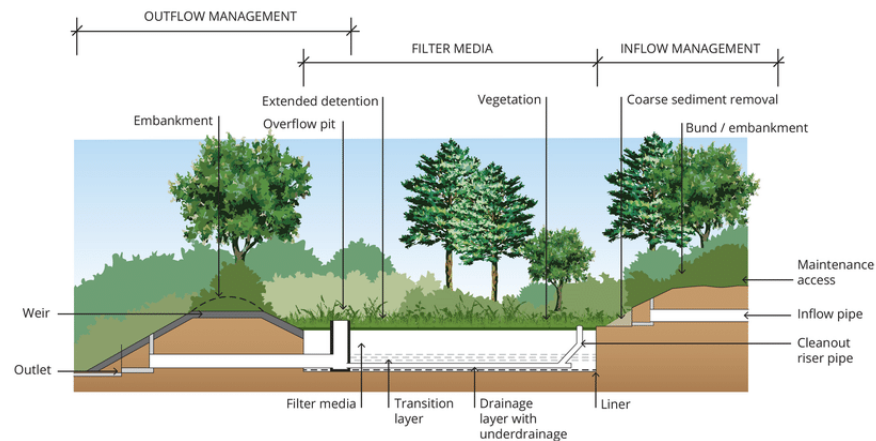


Figure 2. Example of rain garden facilities
(Source: Coombes, 2018)

When urbanization began to increase impervious land and thus cause urban flooding, in order to imitate natural bioretention processes that existed in the nature before urbanization, the first raingardens were made in the state of Maryland by Dick Brinker and Larry Coffman in the late 1980s. They had the idea to replace the traditional best management practices (BMP) pond with a bioretention area. This was applied in a residential subdivision and proven to be highly cost-effective (Kassulke, 2003; Dunnet and Clayden, 2007), so raingardens were seen as a future potential type of green infrastructure.

“Green infrastructure” (e.g., vegetated stormwater solutions) as an alternative to “gray” infrastructure (e.g., stormwater pipes) solutions is becoming more common place as a lower cost infrastructure solution to stormwater management (EPA, 2007; NRDC, 2011). Green infrastructure can be used in conjunction with pipe repair and the expansion of stormwater infrastructure capacity in order to reduce overall amounts of stormwater in the stormwater system (Figure 3).



Figure 3. Raingardens as type of green infrastructure
(Source: bilimgenc.tubitak.gov.tr, 2019)

Beyond the scope of capital improvements and regulations, retrofitting patterns of neighborhood development through the implementation of green infrastructure like sustainable stormwater management facilities is a localized urban planning and design strategy that may also contribute to sustainability and resilience goals. Such goals are addressed through their being a decentralized approach to urban planning that reduces downstream environmental impacts (Newman and Jennings, 2008; Novotny and Ahern, 2010), through increased biodiversity due to expanded habitat (Beatley, 2011), through reduced stormwater runoff (City of Portland, 2010; NRDC, 2011), and through the contribution to groundwater recharge and the water cycle as a whole (Girling and Kellett, 2005). There are some that contend that in addition to these ecological benefits, there are further social benefits of sustainable stormwater infrastructure such as increased access to nature and opportunity to learn about the stormwater system. For example, one researcher, Kathleen Wolf, includes sustainable stormwater infrastructure in a list of “civic nature” that has potential benefits for human health (Wolf, 2008). Indeed, some scholars contend that nature as part of daily life has the potential to build awareness of nature and natural processes which might eventually translate to a love of nature and a desire to protect both the local and global environment (Jordan III, 2000). Although there are studies that explore public acceptance of ecologically designed landscapes, including those that infiltrate stormwater (Canfield and Gibson, 2013), there are few studies that deal specifically with whether sustainable stormwater infrastructure are perceived of as nature by community members (Apostolaki and Jefferies, 2005). Yet another potential benefit of sustainable stormwater infrastructure is due to its visibility. Within the landscape architecture literature, there is a discussion on linking human values to ecology through the design of landscapes that incorporate visible eco-logical processes. Such visibility would allow for observation of ecological processes, which over time might become a valued and normal aspect of the lived environment (Gobster et al., 2007). Moreover, there is an argument that the visibility of waste or resources can contribute to public understanding of the consequences of production and consumption (Hough, 2004; Newman and Jennings, 2008), which thereby might have an effect on individual behaviors (e.g., driving less, reducing the use of fertilizer, or planting native vegetation). Although some scholars specify utilizing educational elements as part of ecological landscapes in order to foster understanding of ecological processes and acceptance

of ecological design (Echols and Pennypacker, 2008; Gobster et al., 2007), this literature review found no research that looks specifically at relationships between stormwater infrastructure visibility and environmental knowledge.

Although aquatic plants are easily chosen for pond and lake design, the choice of plants for raingardens is much more difficult because the length and period of flooding oscillates. Raingardens are neither wet or dry. In most temperate climate areas, among which is the City of Novi Sad, it is unlikely for the raingarden to dry out completely or to retain the water and form a pond, so it is best to choose plants that have the ability to survive in flood and drought conditions, usually those being found around water bodies or in areas of moist soil, or from habitats that are under the influence of significant amounts of rainfall during part of the year (Dunnet and Clayden, 2007). Plants that tolerate drought and flooding are planted mainly in two zones: the zone that is constantly under water and the zone that is dry (Figure 4). Based on that, before planting, plants are selected for each zone. Shrubs, perennials, ground covers are mostly planted in the dry zone, while plants for the flood zone should also have a certain level of drought tolerance. In addition to these two zones, rain gardens also have a third, central zone that can withstand occasional water oscillations whereit is necessary to choose the most resistant plants, such as lotus and some types of ornamental grasses. In general, rain gardens should be designed to filter water for 24 hours (Austin, 2012).

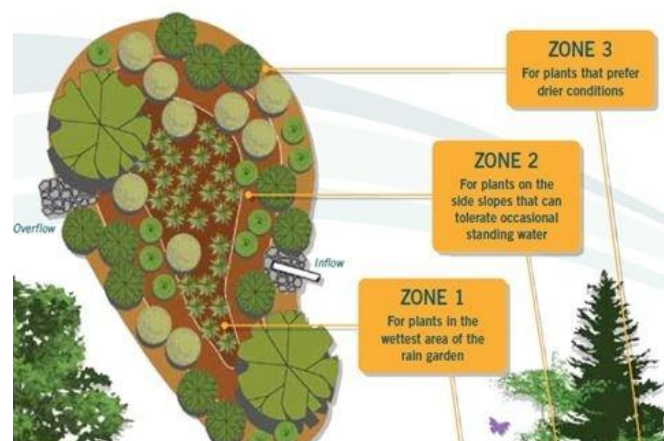


Figure 4. Zones of raingardens

(Source:

<https://www.austintexas.gov/sites/default/files/files/Watershed/growgreen/2019LPT/Rain-Garden-Plants-and-Maintenance-Susan-Kenzle-Darcy-Nuffer.pdf>)

Since many raingardens are located along the roads it is recommended to use plants that are tolerant to salt and have fibrous root system and avoid invasive plants which could damage the underground pipes. In order to enhance the biodiversity and to give the landscape a natural and wild appearance, indigenous plants are recommended because of their resistance and adaptation to existing conditions (Shaw, 2003).

With the right plant selection raingardens would not only fulfill their function, but would contribute to the aesthetic value of the green area. Plants for rain gardens can be grouped into perennials, ground covers, ornamental grasses, ferns and, less often, shrubs and trees of

smaller dimensions. In the first zone the following perennials and ornamental grasses are recommended: *Coreopsis lanceolata* L., *Echinacea purpurea* Moench., *Liatris spicata* Willd., *Iris* sp., *Panicum virgatum* L., *Miscanthus sinensis* Andersson. In the second zone the following perennials and ornamental grasses are recommended: *Cyperus longus* L., *Lythrum salicaria* L., *Solidago speciosa* Nutt., *Panicum virgatum* L., *Festuca* sp., *Polystichum acrostichoide* sSchott., *Miscanthus sinensis* Andersson., *Acorus calamus* 'Variegatus'. In addition to these previously mentioned species the following indigenous plants for Serbia can be used for raingardens: *Ajuga reptans* L., *Cerastium tomentosum* L., *Geranium macrorrhizum* L., *Geumcoccineum* Sm., *Geumurbanum* L., *Hedera helix* L., *Hypericum calycinum* L., *Iberis sempervirens* L., *Iris germanica* L., *Iris pseudacorus* L., *Iris pumila* L., *Iris sibirica* L., *Ruta graveolens* L., *Santolina chamaecyparissus* L., *Tanacetum parthenium* (L.) Sch. Bip., *Veronica spicata* L., *Veronica spicata* L., *Vinca major* L. (Popović and Marković, 2012).

As the final result of research, the principals of raingardens design were applied on the field, on a high-risk flood area (Figure 5). By revitalizing this area, with the careful design (Figure 6) and use of indigenous species listed above, this raingarden would infiltrate and eliminate run-off, absorb pollutants, enhance the biodiversity and add an aesthetic value to the location.



Figure 5. Problems of flooding in Partizanska street
(Source: <https://www.mojnovisad.com/vesti/novosadske-ulice-pod-vodom-dezurne-ekipe-na-terenu-foto-id22678.html>)



Figure 6. 3D of a selected location for a rain garden with plant selection
(Source: Authors, 2021)

CONCLUSION

Sustainable stormwater management facilities such as rain gardens are one way in which cities are simultaneously addressing the need to replace or repair stormwater infrastructure while also meeting regulatory obligations. This article explored the premise that sustainable stormwater infrastructure can be seen as small scale instances of nature, providing means for urban residents to connect with nature and learn about the local stormwater system. Implementing the first raingarden along the road in Novi Sad and later using this model of revitalization for similar high-risk flood areas, would enhance the current green infrastructure of the city and mitigate the frequent problem of urban flooding which will help Novi Sad in creating a “green” and sustainable city.

REFERENCES

- APOSTOLAKI, S. and JEFFERIES, C. (2005): The social impacts of stormwater management techniques. In 10th international conference of urban drainage Copenhagen, Denmark.
- AUSTIN, G. (2012): Design and performance of bioretention beds for removal of stormwater contaminants. *Journal of Green Building*: 17-27.
- BEATLEY, T. (2011): *Biophilic cities: Integrating nature into urban design and planning*. Washington, DC: Island Press.
- CANFIELD, J. L., and GIBSON, H. J. (2013) : Resident attitudes toward Stapleton’s stormwater parks. In Paper presented at the CELA 2013 conference Austin, TX.
- CITY OF PORTLAND: (2010). 2010 stormwater management facility monitoring report summary. Retrieved from. <http://www.portlandonline.com/bes/index.cfm?c=36055&a=343463>
- DIETZ, M.E. (2007): Low impact development practices: A review of current research and recommendations for future directions. *Water, air, and soil pollution*. 186(1-4): 351-363.
- DUNNETT, N., CLAYDEN, A. (2007): *Rain gardens. Managing water sustainably in the garden and designed landscape*. Timber press, 139p.

- DURAKOVIĆ, A., SEKULIĆ M. (2017): Greenstreetsasamodelofstormwater managementin Novi Sad. 5th International conference Contemporary achievements in civil engineering, 21st of April, Subotica, Serbia: 823-832.
- ECHOLS, S., and PENNYPACKER, E. (2008): Stormwater management to artful rainwater design. *Landscape Journal*, 27(2): 268–290.
- EPA. (2007): Reducing stormwater costs through low impact development (LID) strategies and practices. In EPA 841-F-07-006. Washington, DC: United States Environmental Protection Agency. Retrieved from <http://water.epa.gov/polwaste/green/costs07/index.cfm>.
- GIRLING, C., & KELLETT, R. (2005): *Skinny streets and green neighborhoods: Design for environment and community*. Washington, DC: Island Press.
- GOBSTER, P., NASSAUER, J., DANIEL, T., and FRY, G. (2007): The shared landscape: What does aesthetics have to do with ecology? *Landscape Ecology*, 22(7), 959–972.
- GREKSA, A., GRABIĆ, J. and BLAGOJEVIĆ, B. (2021): Contribution of low impact development practices-bioretention systems towards urban flood resilience: case study of Novi Sad, Serbia. *Environmental Engineering Research*, 27(4): 106-117.
- HOUGH, M. (2004): *Cities and natural process: A basis for sustainability*. Routledge.
- ISHIMATSU, K., ITO, K., MITANI, Y., TANAKA, Y., SUGAHARA, T. and NAKA, Y. (2017): Use of rain gardens for stormwater management in urban design and planning. *Landscape and Ecological Engineering*, 13(1): 205-212.
- JORDAN, W. R., III. (2000): Restoration, Community, and Wilderness. In P. Gobster, & B. Hull (Eds.), *Restoring nature: Perspectives from the social sciences and humanities* (pp. 23–36). Washington, DC: Island Press.
- KASSULKE, N. (2003): Rain Gardens Made One Maryland Community Famous. *Wisconsin Natural Resources Magazine*.
- NEWMAN, P., and JENNINGS, I. (2008): *Cities as sustainable ecosystems: Principles and practices*. Washington, DC: Island Press.
- NOVOTNY, V. & AHERN, J. (2010): *Planning and design for sustainable and resilient cities: Theories, strategies, and best practices for green infrastructure*.
- NRDC. (2011): *Rooftops to Rivers II: Green strategies for controlling stormwater and combined sewer overflows*. Natural Resources Defense Council.
- POPOVIĆ, M., MARKOVIĆ, M. (2012): Floristic analysis of perennial species on flowerbeds in Belgrade with special attention on invasiveness of the recorded species. *Glasnik Šumarskog fakulteta*. (106): 169-182.
- SHARMA, R., and MALAVIYA, P. (2021): Management of stormwater pollution using green infrastructure: The role of rain gardens. *Wiley Interdisciplinary Reviews: Water*, 8(2), e1507.
- SHAW, D. B. (2003): *Plants for Stormwater Design: Species selection for the upper midwest* (Vol. 1). Minnesota Pollution Control Agency.
- WANG, L., J. LYONS, P. KANEHL, and R. BANNERMAN. (2001): Impacts of Urbanization on Stream Habitat and Fish Across Multiple Spatial Scales. *Environmental Management* 28(2): 255-266.
- WOLF, K. L. (2008): With plants in mind: Social benefits of civic nature. *Master Gardener*, 2(1), 7–11.

EFFECT OF DAY LENGTH AND EXOGENOUS MELATONIN ON GROWTH HORMONE SECRETION IN SHEEP DURING PREGNANCY AND LAMB REARING

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Summary: *The aim of the research was to investigate the effect of the length of the day and exogenous melatonin on the secretion of growth hormone in sheep during pregnancy and lamb rearing. 60 Polish long-haired sheep ewes divided into 3 groups were used for the experiment. The first blood sampling was performed 6 weeks before the lambing. Then, the sheep from the third group were implanted subcutaneously with melatonin. The second blood sampling was performed on the 57th day of lactation. The introduction of melatonin implants in the perinatal period contributed to the modulation of the melatonin and growth hormone signal in the lambs weaning period.*

Key words: *sheep, day length, melatonin, growth hormone*

INTRODUCTION

The seasonality of the sheep's reproductive cycle depends on the season of the year and the changing length of the day, as well as the organism's internal endogenous signals. Melatonin (MTL), as a biochemical synchronizer of physiological rhythms, modulates reproductive processes in sheep. Endogenous mediators of the effect of day length on milk production may also be some somatotrophic hormones such as growth hormone (GH) (Molik et al., 2013). The action of growth hormone plays an important role in the regulation of several physiological processes, such as growth, lactation and other metabolic processes (Gluckman et al., 1987). The observations so far show that the process of initiating and maintaining lactation is a process dependent on many factors, and changes in the length of the day and changes in the profile of melatonin and growth hormone are the only of many (Molik et al., 2009). Therefore, the aim of the research was to investigate the effect of the length of the day and exogenous melatonin on the secretion of growth hormone and melatonin in sheep during pregnancy and lamb rearing.

MATERIAL AND METHODS

60 Polish long-haired sheep ewes were used for the experiment, which were similar in terms of body weight ($60 \text{ kg} \pm 5 \text{ kg}$) and age (4 - 5 lactation). Sheep were divided into three groups of 20 each. The estrus cycle of females was synchronized using the Chrono-Gest method. The heat time was additionally verified with a ram. Breeding of the first group took place from 10 to 15 September, and lambing took place in mid-February. The breeding period of the second and third groups took place from 10 to 15 January, the lambing and the start of lactation took place in mid-June. Rearing of lambs with their mothers lasted 56 days. The mothers' nutrition was adjusted to their current physiological state. The first blood sampling was performed 6 weeks before the lambing. Later, melatonin implants (Ceva Animal – Melovine 18 mg) were introduced subcutaneously to sheep from group three. Subsequent implants with exogenous melatonin were introduced every 90 days. The second blood sampling was performed on the 57th day of lactation. Blood sampling began at sunset and continued for 6 hours every 20 minutes. Melatonin concentration was determined in 500 μl of whole plasma by the radioimmunoassay (RIA). Determination of the concentration of growth hormone in the blood plasma was performed using the RIA method. All determinations were made at the Department of Endocrinology at the Institute of Animal Physiology and Nutrition of the Polish Academy of Sciences in Jabłonna. The Mann-Whitney test was used to determine the differences in lactation parameters between the groups. Changes in hormone concentration were analyzed using one-way analysis of variance (ANOVA) using the Scheffes test.

RESULTS AND DISCUSSION

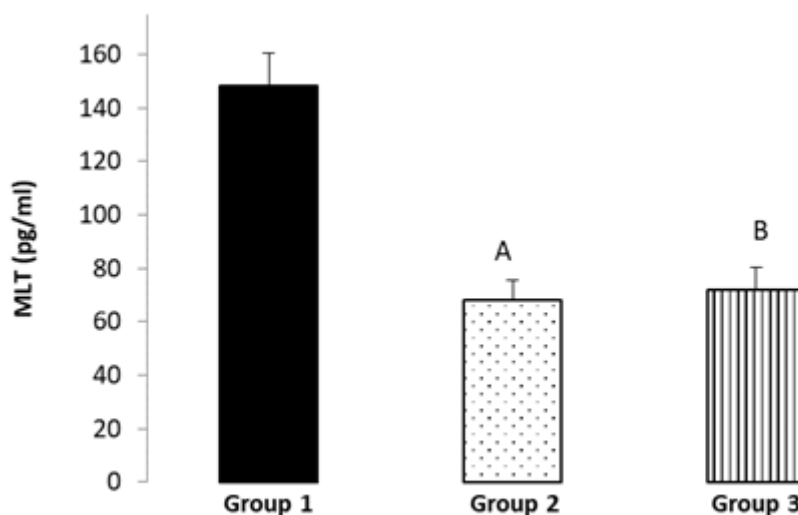
The obtained results indicate a significant influence of the length of the day on the concentration of melatonin in sheep. The analyzes showed that 6 weeks before lambing, the melatonin level was significantly higher ($P \leq 0.01$) in the ewes of group 1 compared to the concentration of melatonin in ewes from group 2 and group 3. The concentration of melatonin in sheep from group 1 was ($148.15 \pm 12.4 \text{ pg/ml}$), while in ewes from groups 2 and 3 it was ($68.0 \pm 7.6 \text{ pg/ml}$ and $72.0 \pm 8.40 \text{ pg/ml}$) (Table 1, diagram 1).

Table 1. Changes in melatonin concentration (MLT) in the tested groups of sheep 6 weeks before lambing (group 1 - January, group 2 and group 3 - May).

Breed of sheep Polish long-haired sheep	Melatonin concentration (pg/ml)	
	$\bar{X}$	SD
group 1 - January (6 weeks before lambing) lambing in February	148,15 ^{A,B}	12,4
group 2 - May (6 weeks before lambing) lambing in June	68,0 ^A	7,6

group 3 - May (6 weeks before lambing) lambing in June + exogenous melatonin	72,0 ^B	8,4
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Diagram 1. Changes in melatonin (MLT) concentration in the tested groups of sheep 6 weeks before lambing (group 1 - January, group 2 and group 3 - May).



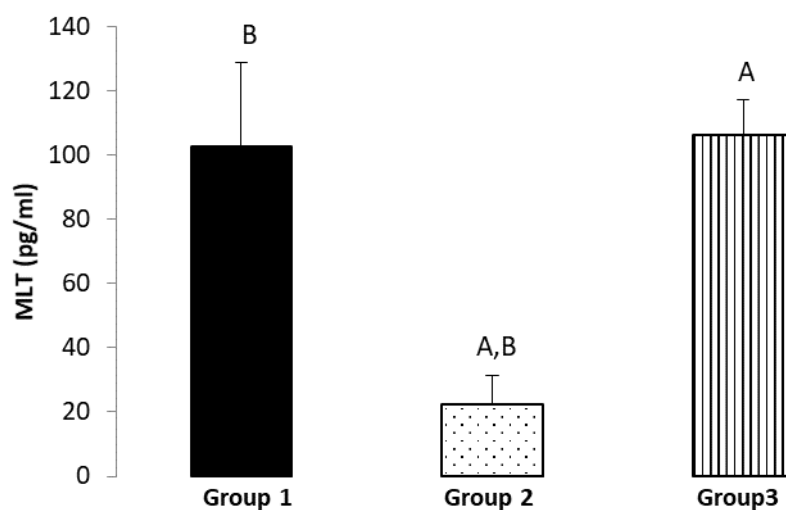
A, B – means marked with the same letters differ significantly with $P \leq 0,01$

The study of the melatonin profile on the 57th day of lactation after weaning the lambs showed that the concentration of melatonin in group 1 was $(102.8 \pm 26.1 \text{ pg/ml})$. Significantly ($P \leq 0.01$) the lowest melatonin concentration was found in sheep from group 2 $(22.4 \pm 8.75 \text{ pg/ml})$. Melatonin concentration in sheep from group 3 was significantly the highest and amounted to $(106.4 \pm 10.7 \text{ pg/ml})$ (Table 2, diagram 2).

Table 2. Changes in melatonin concentration (MLT) in the tested groups of sheep on the 57th day of lactation (group 1 - April, group 2 and group 3 - August).

Breed of sheep Polish long-haired sheep	Melatonin concentration (pg/ml)	
	$\bar{X}$	SD
group 1 - April (57th day of lactation) lambing in February	102,8 ^B	26,1
group 2 - August (57th day of lactation) lambing in June	22,44 ^{A,B}	8,75
group 3 - August (57th day of lactation) lambing in June + exogenous melatonin	106,4 ^A	10,7

Diagram 2. Changes in melatonin (MLT) concentration in the tested groups of sheep on the 57th day of lactation (group 1 - April, group 2 and group 3 - August).



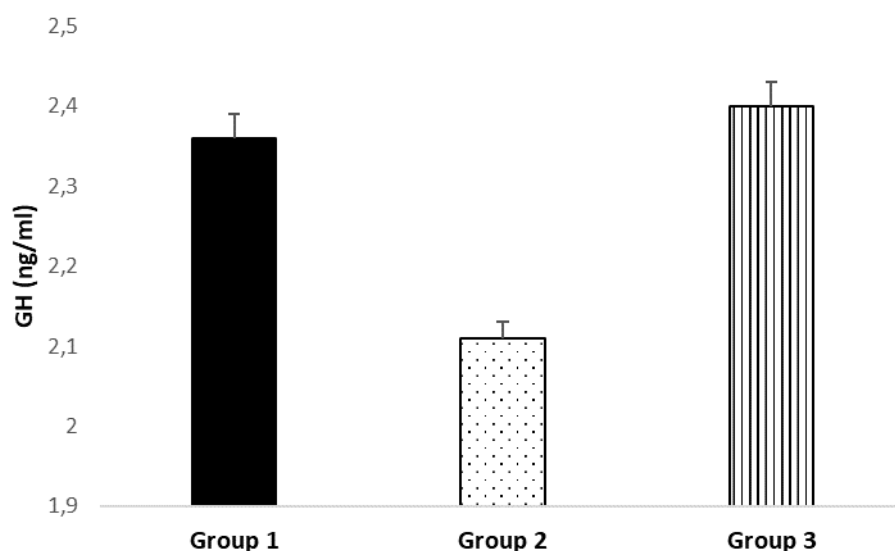
A, B – means marked with the same letters differ significantly with $P \leq 0,01$

Analyzing changes in the concentration of growth hormone in the tested groups of sheep 6 weeks before lambing, it was shown that the highest concentration of GH occurred in sheep from group 3 ($2.40 \pm 0.03 \text{ ng/ml}$), while the lowest in sheep from group 2 ($2.11 \pm 0.02 \text{ ng/ml}$). The concentration of GH in sheep from group 1 was ($2.36 \pm 0.03 \text{ ng/ml}$). There were no statistically significant differences in the concentration of GH in the tested groups of sheep 6 weeks before lambing (Table 3, diagram 3).

Table 3. Average concentrations of growth hormone (GH) in the tested groups of sheep 6 weeks before lambing (group 1 - January, group 2 and group 3 - May).

Breed of sheep Polish long-haired sheep	Growth hormone concentration (ng/ml)	
	$\bar{X}$	SD
Group 1 - January (6 weeks before lambing)	2,36	0,03
Group 1 - May (6 weeks before lambing)	2,11	0,02
Group 1 - May (6 weeks before lambing)	2,4	0,03

Diagram 3. Average concentrations of Growth hormone (GH) in the tested groups of sheep 6 weeks before lambing (group 1 - January, group 2 and group 3 - May).

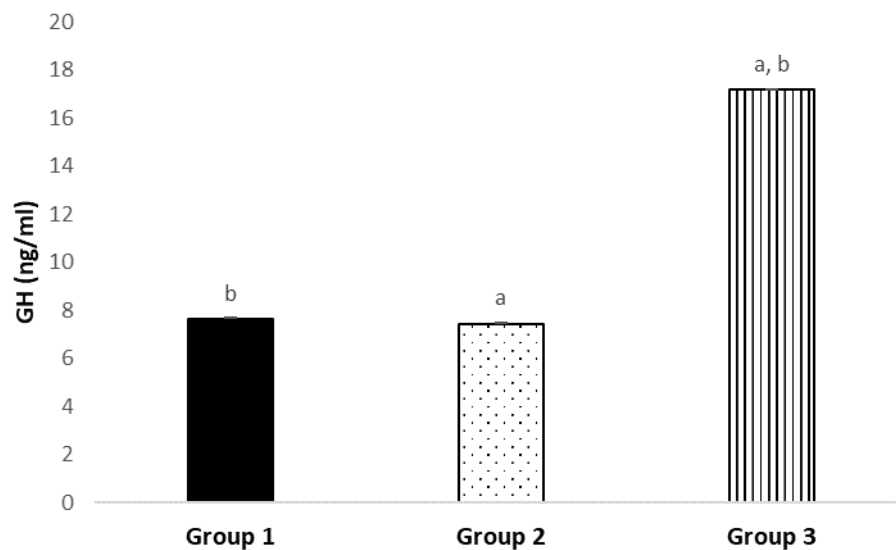


Analyzing changes in the concentration of growth hormone in the tested groups of sheep on the 57th day of lactation, it was shown that the highest concentration of this hormone was in animals from group 3 ($8.73 \pm 0.04 \text{ ng/ml}$), while significantly ($P \pm 0.05$) the lowest was in sheep group 2 ($7.44 \pm 0.04 \text{ ng/ml}$) and group 1 ($7.64 \pm 0.04 \text{ ng/ml}$) (Table 4, diagram 4).

Table 4. Average concentrations of growth hormone (GH) in the tested groups of sheep on the 57th day of lactation (Group 1 - April, Group 2 and Group 3 - August).

Breed of sheep Polish long-haired sheep	Growth hormone concentration (ng/ml)	
	$\bar{X}$	SD
Group 1 - April (57th day of lactation)	7,64 ^b	0,04
Group 2 - August (57th day of lactation)	7,44 ^a	0,04
Group 3 - August (57th day of lactation)	17,17 ^{a,b}	0,04

Diagram 4. Average concentrations of growth hormone (GH) in the tested groups of sheep on the 57th day of lactation (Group 1 - April, Group 2 and Group 3 - August).



a,b – means marked with the same letters differ significantly $P \leq 0,05$

At the 6th week before lambing, the sheep from group 1 had the significantly highest levels of melatonin compared to groups 2 and 3. The results obtained from the studies carried out on sheep on the 57th day of lactation (after weaning the lambs) show that the ewes from group 3 had the highest melatonin concentration. Stimulating the short-day signal by administering exogenous melatonin and shortening the length of the day, cause oestrus symptoms due to an increase in the level of luteinizing hormone (LH) (Misztal and Molik 2005; Molik et al., 2013). It has been proven that the length of the day, and therefore melatonin, has a significant influence on the lactation efficiency of sheep. In lactating sheep, GH secretion increases significantly during suckling, especially during the first 10–30 minutes (Misztal et al., 2008). In sheep, the short-day period results in increased pineal secretory activity, high MLT concentration, low PRL level, which indirectly inhibits lactation (Misztal and Molik, 2005).

CONCLUSION

Studies on changes in melatonin secretion 6 weeks before lambing showed a significantly higher concentration of melatonin in sheep from group 1 compared to sheep from group 2. The results obtained on the 57th day of lactation showed a significantly lower concentration of melatonin in sheep from group 2 compared to sheep from group 3. Growth hormone concentration was the highest in group 3, and the lowest in group 2, both 6 weeks before lambing and on the 57th day of lactation. The introduction of melatonin implants in the perinatal period contributed to the modulation of the melatonin and growth hormone signal in the lambs weaning period.

REFERENCES

- GLUCKMAN P.D., BREIER B.H., DAVIS S.R (1987): Physiology of the somatotrophic axis with particular reference to the ruminant. *Journal of Dairy Science*, 70: 442–466.
- MISZTAL T., MOLIĆ E. (2005): Aktualny stan wiedzy na temat mechanizmów kontrolujących aktywność płciową i laktację u owiec. *Biologiczne i kulturowe aspekty gospodarki owczarskiej*, 51 – 59.
- MISZTAL T., GÓRSKI K., TOMASZEWSKA-ZAREMBA D., MOLIĆ E., ROMANOWICZ K. (2008); Identification of salsolinol in the mediobasal hypothalamus of lactating ewes and its relation to suckling-induced prolactin and GH release. *Journal of Endocrinol*, 198; 83-89.
- MOLIĆ E., KALISZ B., MATYJA A. (2009); Wpływ długości dnia na zmiany sekrecji melatoniny i prolaktyny u owiec. *Wiadomości Zootechniczne*, R. XLVII 2: 13–17.
- MOLIĆ E., PASTERNAK M., BŁASIAK M., MISZTAL T., ROMANOWICZ K., ZIEBA D. (2013); The effect of the diversified signal of melatonin on milk yields in seasonally breeding sheep. *Archives Animal Breeding*, 56, 924-932.
- MOLIĆ E., MISZTAL T., ROMANOWICZ K., ZIEBA D. (2013); Short-day and melatonin effects on milking parameters, prolactin profiles and growth-hormone secretion in lactating sheep, *Small Ruminant Research*, 109(2-3).

INFLUENCE OF YEASTS ON THE CHEMICAL COMPOSITION AND QUALITY OF GRAŠEVINA WINES FROM CLONES OB-412 AND OB-445

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Summary: *Graševina* is the most common grapevine variety in Croatian vineyards. Since 2004, it was the object of clonal selection which has resulted in four certified clones. In this research, which includes grapes from clones OB-412 and OB-445, the fermentations were carried out with different strains of *Saccharomyces cerevisiae* yeast and sequential inoculation with *Metschnikowia pulcherrima* and *Saccharomyces cerevisiae*. The basic physicochemical parameters of must and wine were analyzed as well as individual organic acids and total phenols. With significant differences among individual parameters, samples inoculated with *S. cerevisiae* strains were better sensory evaluated.

Key words: *Graševina*, *Metschnikowia pulcherrima*, non-*Saccharomyces* sp, sequential fermentation

INTRODUCTION

Graševina is a white grape variety which is cultivated in most vineyards of Continental Croatia, in a total of 4.563,62 ha (APPRRR, 2019). The range of wine styles produced by this variety is wide, from simpler dry and fresh to sweet predicate and more complex wines. Because Graševina is an extremely important variety in Croatia in 2004 it became the object of clonal selection. The goal was to select, protect and propagate different genotypes of the Graševina variety. The process resulted in four registered clones that stood out through ampelographic evaluation in yield, average cluster weight, sugar and total acidity as well as the must pH values (Preiner, 2018). The clones also went through enological evaluation. Wines from all four clones were produced so that their chemical composition and quality could be evaluated.

The aim of this research was to determine differences in basic physicochemical and sensory characteristics of Graševina wines considering the combination of selected yeasts that carried out alcoholic fermentation. Clone OB-445 has average quality indicators and it is intended for the production of simpler, but typical wines of this variety with slightly lower alcohol content and less pronounced body. Clone OB-412 gives a wine that has a fuller body and higher alcohol content, but at the same time harmonious due to higher acid content.

The most used yeast species in winemaking is *Saccharomyces cerevisiae* (Sc), regarding its ability to complete alcoholic fermentation. Non-*Saccharomyces* yeasts, such as

Metschnikowia pulcherrima (*Mp*), until recently have been considered negative and undesirable triggers for spontaneous fermentation with uncertain termination. However, in the last 15 years some species have been selected and commercially available because of the positive impact they have on wine properties. Their application is possible in sequential fermentation with compatible *Sc* strains due to their lower fermentable potential. The ability of *Mp* to release higher concentrations of varietal flavors and lower ethanol concentration in wines has stimulated interest in its commercial use (Ivitić *et al.*, 2020).

MATERIAL AND METHODS

The research was made in harvest 2020, at the experimental viticultural and enology station Jazbina owned by the Faculty of Agriculture in Zagreb. The grapes of both clones were hand-picked, quickly processed, and after 24 hours of settling, each must was divided into three treatments with two replications which meant six 15 L-glass containers. Control treatment (C) OB-412 and OB-445 were inoculated with *S. cerevisiae* strain (Lalvin Sensy™, Lallemant), treatment (S) OB-412 and OB-445 with *S. cerevisiae* (Sauvay™, Lallemant) and the third treatment (SF) OB-412 and OB-445 was sequentially inoculated with *Metschnikowia pulcherrima* (Flavia™, Lallemant) and *S. cerevisiae* (Sauvay™, Lallemant) after 24 h. Yeast nutrition was used in all treatments according the producers instruction (Stimula™, Lallemant).

The sugar concentration in the must was determined by a refractometer and expressed in Oechsle (°Oe) degrees. Reducing sugar in the wine was analyzed by the Rebelein method (Zoecklein *et al.* 1995), the total acidity (as tartaric, g/L) of must and wine was analyzed by sample neutralization with 0.1 M NaOH with bromothymol-blue indicator according to the OIV method (2012). The alcoholic strength was analyzed by the distillation method based on the specific weight of the distillate at 20°C. Using the Reichardt tables, the alcohol concentration was determined in vol. %. The total phenols in the wine were analyzed by spectrophotometry based on Singleton and Rossi (1965). The method is based on the color reaction of phenolic compounds with Folin-Ciocalteu reagent.

Sensory evaluation was performed by ranking and descriptive method. The descriptive method is used to describe individual sensory properties of wine, and numerically the intensity of individual properties. In the ranking method, the samples must be ranked in order of quality from the best to the worst (Herjavec *et al.*, 2007).

Statistical analysis was performed using XLSTAT software v.2020.3.1. (Addinsoft, New York, NY, USA) using t-test. Results of statistical analysis were considered significant when $p < 0.05$.

RESULTS AND DISCUSSION

Basic psihycochemical composition of must

The basic composition was analyzed in must obtained from clones OB-445 and OB-412. The sugar concentration in must from the clone OB-445 was 89 °Oe and 87 °Oe in must from clone OB-412. By measuring the total acidity, it was confirmed that the must OB-445 contains 7.5 gL⁻¹ of total acids and the must OB-412, 8.1 gL⁻¹. The clone OB-412 had a higher total acidity while the clone OB-445 had a higher sugar concentration which usually means higher concentration of the alcohol later on.

Grapes contains organic acids, the most important ones are tartaric, malic and citric acid. The tartaric acid was presented in must OB-445 with 6.8 gL⁻¹ and 7.24 gL⁻¹ in must OB-412. Following acid was malic acid, in must OB-445 with 0.92 gL⁻¹ and in must OB-412 with 0.83 gL⁻¹. Citric acid had the lowest concentration, respectively 0.08 gL⁻¹ and 0.01 gL⁻¹ in clones OB-445 and OB-412.

Basic physicochemical composition of wine

Table 1. Basic physicochemical composition of Graševina wine clone OB-445

Parameter	(K) OB-445	(S) OB-445	(SF) OB-445	Average
Alcohol (vol%)	13.1 ^a	13.1 ^a	12.7 ^b	12.9
Total extract (gL ⁻¹)	22.2 ^a	21.4 ^b	19.6 ^c	21.1
Reducing sugar (gL ⁻¹)	1.3 ^a	1.6 ^a	1.5 ^a	1.5
Total acidity(gL ⁻¹)	7.0 ^a	7.2 ^a	7.0 ^a	7.1
Volatile acidity (gL ⁻¹)	0.32 ^a	0.24 ^b	0.46 ^a	0.34
pH	2.82 ^b	2.81 ^b	3.00 ^a	2.88
Ash (gL ⁻¹)	2.03 ^b	2.15 ^a	1.30 ^c	1.83

All data present mean value (n=2); Different lower case superscript letters in the same row represent statistically significant differences between means at p≤0.05.

Table 1. shows the basic compositions of clone wines OB-445 in three treatments. The concentration of reducing sugar was an indication that alcoholic fermentation was completed (< 4 gL⁻¹) and all wines were in the category of dry wines. The alcoholic strength of wine SF OB-445 was significantly the lowest (12.7 vol%) which can be attributed to the influence of *Metschnikowia pulcherrima*, which has the ability to lower ethanol concentrations in wine (Röcker *et al.* 2016). The volatile acidity in the wine S OB-445 was significantly the lowest with 0.24 gL⁻¹. No significant differences were observed in total acidity. Wines from clone OB-445 were characterized by an extremely low pH value resulting from concentrations of organic acids. The highest pH value was noticed in wines (SF) OB-445. Wines of both clones

produced by sequential fermentation (SF) significantly stood out with the lowest concentration of ash and extract, which can negatively affect quality of the wine.

Table 2. Basic physicochemical composition of Graševina wine clone OB-412

Parameter	(K) OB-412	(S) OB-412	(SF) OB-412	Average
Alcohol (vol%)	12.8 ^b	13.0 ^a	12.7 ^b	12.8
Total extract (gL ⁻¹)	20.9 ^b	22.4 ^a	19.0 ^b	20.8
Sugar reducing (gL ⁻¹)	< 1.0 ^a	< 1.0 ^a	< 1.0 ^a	< 1.0
Total acidity (gL ⁻¹)	7.1 ^a	6.7 ^b	6.3 ^c	6.7
Volatile acidity (gL ⁻¹)	0.41 ^a	0.32 ^b	0.42 ^a	0.38
pH	2.82 ^b	2.81 ^b	3.05 ^a	2.89
Ash (gL ⁻¹)	1.64 ^b	1.96 ^a	1.36 ^c	1.65

All data present mean value (n=2); Different lower case superscript letters in the same row represent statistically significant differences between means at $p \leq 0.05$

Table 2. shows the results of the physicochemical analysis of OB-412 wines. The concentration of reducing sugar in all three wines was less than 1 gL⁻¹, which confirms that fermentation was completely over and all wines were dry.

The alcoholic strength was significantly lower in (K) and (SF) OB-412 wines. The (SF) OB-412 wine had significantly the lowest concentration of total acids (6.3 gL⁻¹). The wine S OB-412 also had a significantly lowest concentration of volatile acids compared to other treatments. The (SF) OB-412 sample, produced by sequential fermentation, had significantly the highest pH value.

Individual organic acids and total phenols

After the fermentation was completed concentrations of organic acids have changed. The concentration of tartaric acid in both clones was the lowest in wines obtained by sequential fermentation, manifested through higher pH value of wine (Fig. 1). Also, the concentration of malic acid decreased and the lowest concentration was noticed in the wine (S) OB-412 and the highest in the wine (SF) OB-445. The highest concentration of citric acid was in the wine (SF) OB412. Lactic and succinic acid are most often produced during fermentation by the activity of yeast and bacteria. In the wine (S) OB-412 lactic acid was not detected, while in the control wines OB-445 and OB-412 the value of 0.13 gL⁻¹ was recorded. The succinic acid concentration was the highest in (K) OB-412 and lowest in wine (K) OB-445.

The highest concentration of total phenols (320 mgL⁻¹) was detected in wine (S) OB-445 obtained with the Sc strain Sauvy TM (Fig. 2). The lowest total phenol concentration of 254 mgL⁻¹ was in wine (K) OB-445 (Lalvin SensyTM).

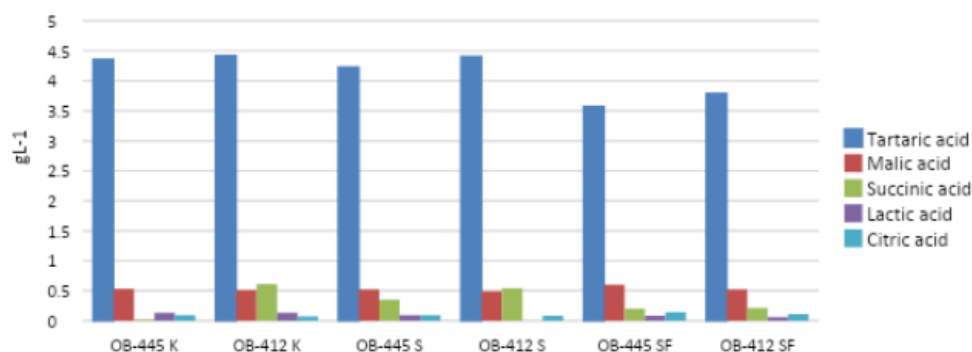


Fig.1 Concentration of organic acids (g L^{-1}) in Graševina wines

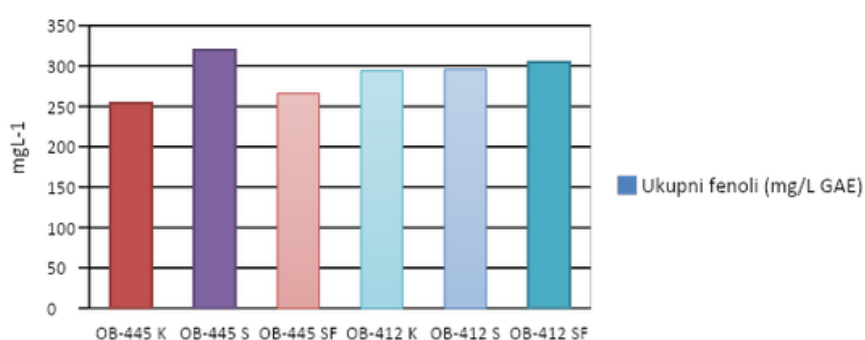


Fig. 2 Concentration of total phenols (mg L^{-1}) in Graševina wines

Sensory analysis of wine

Fig. 3 shows that among clone OB-445 wines, the highest intensity and quality of color was noticed in wine (S) OB-445. The (K) OB-445 wine had the most intense floral aromas, while they were the least pronounced in wine (SF) OB-445. Fruit aromas were more pronounced in (K) OB-445 and (S) OB-445 wines. Wine obtained by sequential fermentation was dominated by intense aromas of spices and herbs. Acidity was more pronounced in (K) OB-445 and (S) OB-445 samples, which correlated with higher concentrations of tartaric acid. The sample (S) OB-445 was rated as wine with the most complex body, and wine (K) OB-445 was rated as the most harmonious sample. *Aftertaste* was the best in the sample (S) OB-445 together with overall impression. The sample (S) OB-445 was the best ranked sample, and the worst was (SF) OB-445.

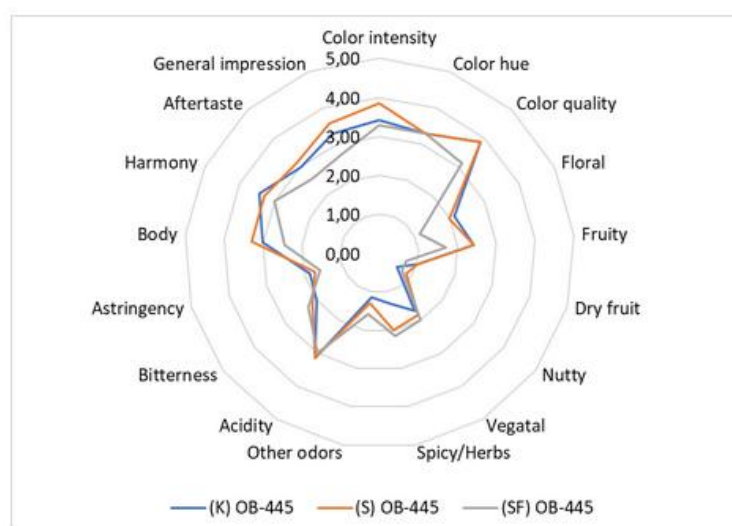


Fig. 3 Descriptive sensory evaluation of Graševina wines from clone OB-445

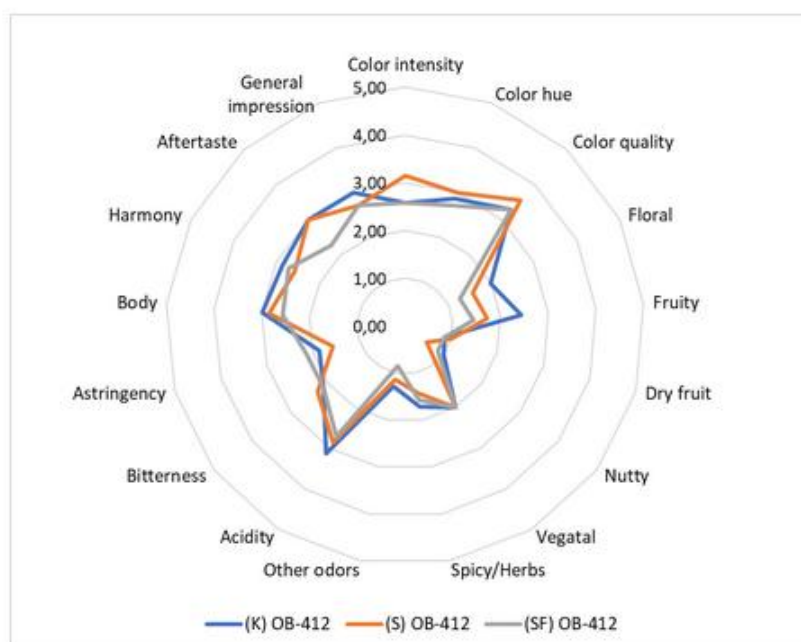


Fig. 4 Descriptive sensory evaluation of Graševina wines from clone OB-412

The best ranked sample from clone OB-412 was control wine (Lalvin Sensy™). The highest color intensity was recorded for wine (S) OB-412. Flower and fruity aromas were more intense in (K) OB-412 wine as well as acidity (Fig. 4). The most complex body was attributed to wine (K) OB-412. *Aftertaste* was the lowest in (SF) OB-412 sample, while the other two samples shared the same scores. The (K) OB-412 sample was also best rated by the overall impression.

CONCLUSION

The aim of the research carried out in the harvest of 2020 was to analyze the effect of different yeast species (*Saccharomyces cerevisiae* i *Metschnikowia pulcherrima*) and their combinations in sequential fermentation, on the chemical composition and quality of Graševina wines made from clones OB-445 and OB-412. In all wine samples, alcoholic fermentation finished completely. According to the results of the basic analysis, a significant difference in the concentration of alcohol was observed in sequential inoculation with *M. pulcherrima* (SF OB-445). The analyses revealed the lowest concentrations of tartaric acid in wines after sequential fermentation, while the concentrations of citric acid were the same. The influence of yeast on total phenols was determined and the highest concentration (320 mgL⁻¹) was in the wine OB-445 fermented with Sauvvy™. After the sensory evaluation, control wine OB-412 was the best rated. Wines OB-445 produced with *S. cerevisiae* were aromatically very similar, while wine obtained by sequential fermentation was marked by nutty and spicy aromas, lower graded body, *aftertaste* and overall impression. Both clone wines produced with Sauvvy™ *Sc* strains were highlighted with a higher quality color.

REFERENCES

- HERJAVEC, S., MARIĆ, J., JEROMEL, A. (2007): Senzorno ocjenjivanje vina. Agronomski fakultet Sveučilišta u Zagrebu.
- IVIT, N. N., LONGO, R., KEMP, B. (2020): The effect of non-Saccharomyces and Saccharomyces non-cerevisiae yeasts on ethanol and glycerol levels in wine. *Fermentation*. 6(3), 77.
- O.I.V. (2012): International code of oenological practices, 01, Paris.
- PREINER, D. (2018): Rezultati završnog ispitivanja klonskih kandidata sorte 'Graševina' predloženih za priznavanje
- RÖCKER, J., STRUB, S., EBERT, K., GROSSMANN, M. (2016): Usage of different aerobic non-Saccharomyces yeasts and experimental conditions as a tool for reducing the potential ethanol content in wines. *European Food Research and Technology*. 242(12), 2051-2070.
- SINGLETON, V. L., ROSSI, J. A. (1965): Colorimetry of total phenolics with phosphomolybdic-phosphotungstic acid reagents. *American Journal of Enology and Viticulture*. 16(3), 144-158.
- ZOECKLEIN, B., FUGELSANG, K.C., GUMP, B.H., NURY, F.S. (1995): Wine analysis and production. New York, Chapman & Hall.
- <https://www.apprrr.hr/>
- <https://www.lallemandwine.com/en/australia/products/catalogue/wine-yeasts/80/lalvin-sensy/>

PROXIMAL SENSING AND VEGETATION INDICES OF WHEAT FOR PREDICTIVE AGRICULTURE

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Summary: *Climate change will have the impact on wheat production, in the form of lower harvests, higher costs, and necessary deviation from traditional farming. Monitoring the wheat canopy using remote and proximal sensors during the growing season plays an important role in site-specific management decisions, but could also improve understanding of the abiotic and physiological processes that control plant N uptake, yield, and protein content. As automated sensors provide information on wheat development, we are exploring the hypothesis that proximal canopy reflectance data could define plant phenology in new ways throughout the season, allowing better prediction of crop yields and detection of plant stress caused by climate change parameters. Spectroradiometers and multispectral imaging are capable of measuring the spectral reflectance of plants, providing vegetation indices that are valuable for assessing the amount and condition of vegetation. This paper presents ongoing work on the use of a system to collect multispectral data on wheat development under different climatic conditions, artificially generated and modified in prototype climate chambers.*

Key words: *climate change, wheat, multispectral imaging, spectroradiometer, prediction*

INTRODUCTION

Wheat is one of the world's most important staple foods and the primary source of calories and plant protein in the human diet (Curtis et al., 2002). Wheat production needs to be increased to meet the growing population and their increasing food demand by the middle of this century. In addition, the effects of climate change, such as less favorable growing conditions, may further reduce wheat production (Rahman et al., 2021). A study by Warrick (1988) for Western Europe, UK and USA on the effects of global warming on wheat productivity shows catastrophic effects in terms of yield loss as higher temperatures accelerate the evapotranspiration process and cause moisture deficit stress. Humid conditions are beneficial for wheat yield while drier conditions are detrimental and reduce productivity. Moreover, Sekar and Pal (2012) have shown that wheat productivity growth has already slowed down in recent decades, which is a major problem for international food security. In this context, appropriate adaptation measures may be needed. Therefore, it is

important to better understand how wheat yield production is changing and how it might be affected by additional stresses due to climate change in the future. Without adaptation measures, climate change could threaten the global supply of wheat.

While grain yield is the primary trait of interest, it can be estimated using proximal or remote sensing data (Lillesand et al., 2014). Proximal sensing technologies could be used to detect the variables in the plant that are responsible for yield variability. The full potential of these technologies for precision agriculture can be exploited in combination with innovative data processing methods, such as machine learning algorithms, to provide useful information for yield management (Abbas et al., 2020), but also to improve the understanding of the abiotic and physiological processes that control nitrogen uptake, yield and protein content in grain (Magney et al., 2016). Recently, several plant phenotyping platforms have been developed to determine the physiological status of plants exposed to stress conditions. The most widely used methods for non-destructive investigation of phenotypic traits of plants under stress conditions are multispectral imaging (Lazarević et al., 2021) and sensors such as spectroradiometers (Carter, 1993). The use of leaf reflectance spectra to predict leaf properties is a promising alternative to conventional measurements because reflectance can be measured quickly and efficiently with the above equipment. A leaf reflectance spectrum results from the light absorbing and scattering properties of the leaf and is greatly modified by the concentration of light absorbing compounds such as photosynthetic pigments, water, sugar, proteins, etc., and the internal structure of the leaves, which determines light scattering. As leaf structure and chemical composition vary with species, age, environment and stress, these light-absorbing and light-scattering properties change together (Burnett et al., 2021). Spectroradiometers and multispectral imaging devices measure the spectral reflectance of plants and therefore provide vegetation indices. Vegetation indices can describe the health and condition of agricultural crops, and each index uses a different part of the electromagnetic spectrum and therefore has diverse informative value (Jelinek et al., 2020). Vegetation indices are often applied to remote sensing data to develop a proxy value for a biophysical variable. Numerous vegetation indices have been developed, and many are reported to correlate with plant condition, stress, vegetation fraction, or chlorophyll content (Gitelson et al., 2002). One of the most commonly used and applied indices, calculated from multispectral information as a normalized ratio between the red and near-infrared bands, is the Normalized Difference Vegetation Index (NDVI). A direct use of NDVI is to characterize canopy growth or vigor. Therefore, it has been compared in many studies with Leaf Area Index (LAI) where LAI is defined as the area of single sided leaves per ground area (Xue and Su, 2017). This paper represents an ongoing work whose main objective is to identify phenological and physiological changes in spectral leaf characteristics of wheat in relation to the effects of climate change. This article is supported by the European Regional Development Fund through the project Advanced and predictive agriculture for resilience to climate change (AgroSPARC) (KK.05.1.1.02.0031).

MATERIALS AND METHODS

Plant Material and Growth Conditions

To initiate the vernalization process of winter wheat, seeds were imbibed on damp paper in a Petri dish for 2 days at 5 °C. The imbibed seeds were then potted in special containers with soil and the seedlings were grown for 8 weeks under short-day conditions (10 hours) at 5 °C in a climate chamber. After this period, the wheat plants were transferred to a prototype climate chamber where they were grown under different artificially created climate conditions - climate change predictions. The proposed method uses the developed encapsulated chamber for experiments on plant growth in an isolated environment with autonomous permutation of artificial climate conditions (light profile, temperature, humidity, air flow and nutrient content) and archiving indicators of plant growth and development detected by spectral sensors (Lešić et al., 2021) (Figure 1).



Figure 1. Prototype climate chamber

The artificially created climatic conditions for wheat cultivation in this study are shown in Table 1.

Table 1. Climatic conditions for cultivation of wheat

Air temperature	25 °C day/18 °C night
CO₂ concentration	350 to 400 ppm
Lighting	500-1500 $\mu\text{mol}/\text{m}^2/\text{s}$ of PPFD
Soil moisture	35 – 45 %

Plant Measurements

Wheat plants were measured at phenophase BBCH 30 – Stem elongation (37 Flag leaf visible).

- Multispectral imaging

Multispectral imaging was performed using the CropReporter™ (PhenoVation B.V., Wageningen, Netherlands). The CropReporter™ consists of a housing with a camera system containing a control computer, a charge-coupled device (CCD) camera with optical filter wheel and focusing unit, and integrated high-intensity red light-emitting diodes (LEDs) for stimulating photosynthesis, LEDs in six spectral bands [broadband white (3000 K), far-red (730 nm), red (660 nm), green (520 nm), blue (460 nm), and UV/blue (405 nm)], controllable in intensity (0-780 $\mu\text{mol m}^{-2} \text{s}^{-1}$), and spectrum for spectral imaging. All images were acquired with the same lens (10 Mp lens, 200 Lp mm^{-1} resolution, 400-1000 nm spectral range) and the same CCD camera (1.3 Mp, 1296 $\times$ 966 pixels) with a true 14-bit signal resolution. Plants were recorded at a distance of 80 cm from the camera. The output is in 16-bit format RAW. Automatic analysis of multispectral images was performed using DATM software (PhenoVation B.V., Wageningen, Netherlands) (Lazarević et al., 2021).

- Spectral reflectance data using spectroradiometer

Spectral reflectance data from wheat leaves were collected using the Spectral Evolution® SR-2500 field spectroradiometer. The instrument has 768 bands covering the entire spectral range from 350 to 2500 nm, with a spectral resolution of 5 nm (350-1000 nm)/22 nm (1500 nm)/22 nm (2100 nm)), together with a contact probe (5 W tungsten halogen lamp). When collecting reflectance data from the wheat leaf samples in the laboratory, the leaves were placed on a black matte paper and the contact probe was placed directly on the sample surface.

PRELIMINARY RESULTS

The wheat was successfully grown under the conditions of the climatic chamber. There were 12 potted wheat plants in phenophase BBCH 30 - Stem elongation (37 flag leaf visible). The study was conducted from July 2021 to November 2021.

Figure 2 shows multispectral images of wheat, where A1 and B1 represent plants taken at a distance of 80 cm from the multispectral camera, and A2 and B2 are macro images of the same wheat plants.

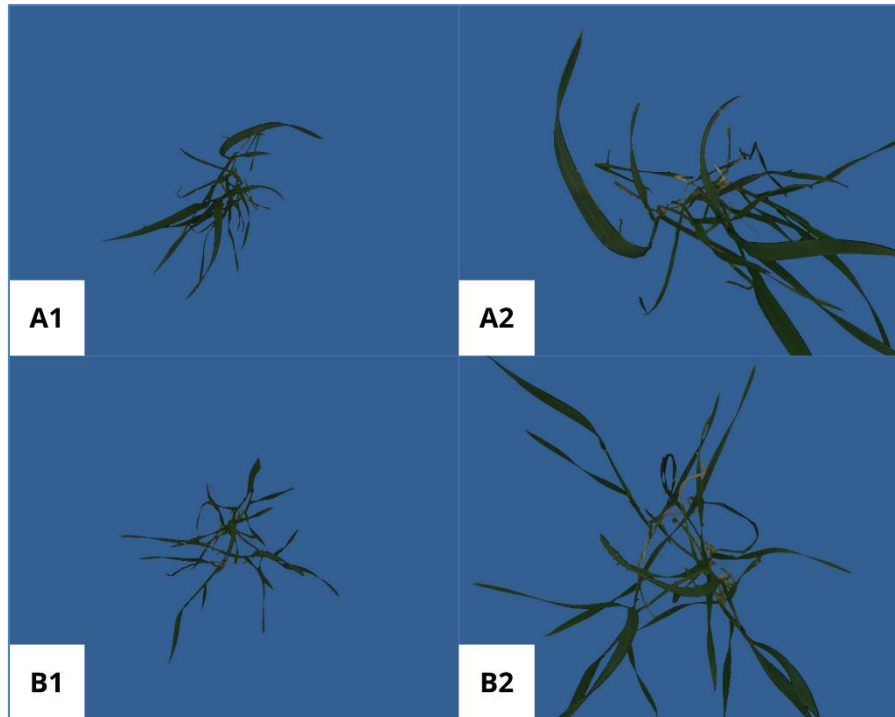


Figure 2. Multispectral imaging of wheat plants

The absolute reflectance values were used to calculate the NDVI ($NDVI = (NIR - Red) / (NIR + Red)$). The average values are shown in Table 2.

Table 2. Average NDVI per plant image

Plant image:	Average NDVI:
A1	0,798125
A2	0,8255
B1	0,698125
B2	0,81725

The same plants were measured with a spectroradiometer. The spectral reflectances recorded with the spectroradiometer in the range from 400 to 2500 nm are shown in Figure 3. The plants differ in the level of reflected radiation over the entire spectral range. However, these differences are most pronounced in the range from 700 nm to 1400 nm. The reflectance pattern showed the same trend for the four leaf samples, but the reflectance of two leaf samples (Wheat 1_00001 and Wheat 2_00002) was higher than the reflectance of the other two leaf samples across the entire spectrum. Those samples were flag leaves of two different wheat plants. The highest spectral reflectance was found in the infrared spectral range (700 - 1300 nm), a relatively low reflectance in the spectral range (1450 - 1800 nm), while the lowest reflectance was found in the spectral range (1950 - 2500 nm). Three spectral wavebands stand out on the spectral print: around 680, 1450 and 1950 nm.

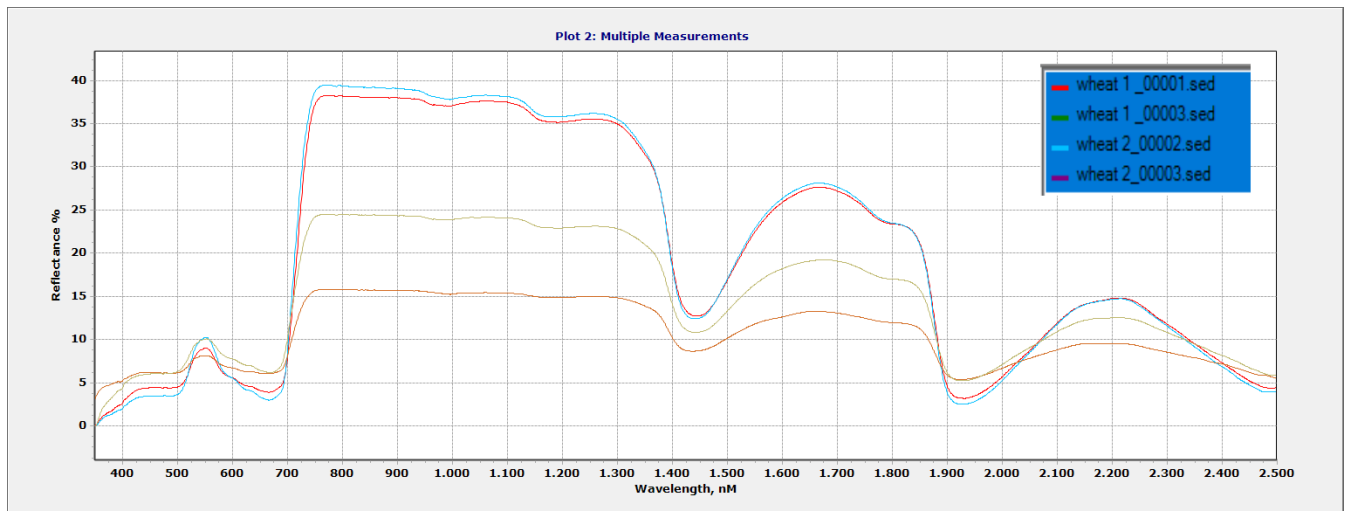


Figure 3. Wheat leaf reflectance data

The absolute reflectance values were used to calculate the NDVI ($NDVI = (NIR - Red) / (NIR + Red)$). The average values are shown in Table 3.

Leaf sample:	NDVI:
Wheat 1_00001	0,799
Wheat 1_00003	0,837
Wheat 2_00002	0,433
Wheat 2_00003	0,576

CONCLUSIONS

Growing wheat in climate chambers with controlled growing conditions cannot mimic field conditions and circumstances, but it provides a good opportunity to control and modify growing conditions such as temperature, moisture, and soil, as well as the duration and intensity of lighting. However, the relationship between soil, plant and atmosphere is different in such systems than in the field. Therefore, careful interpretation of the obtained data on spectral reflectance of wheat leaves is important. The preliminary results show the potential of applying spectral imaging of plants and obtaining a large amount of data in terms of spectral resolution and the possibility of collecting data with high temporal resolution.

LITERATURE

ABBAS, F., AFZAAL, H., FAROOQUE, A. A., AND TANG, S. (2020): Crop yield prediction through proximal sensing and machine learning algorithms. *Agronomy*, 10(7), 1046.

BURNETT, A. C., ANDERSON, J., DAVIDSON, K. J., ELY, K. S., LAMOUR, J., LI, Q., ... AND SERBIN, S. P. (2021): A best-practice guide to predicting plant traits from leaf-level hyperspectral data using partial least squares regression. *Journal of Experimental Botany*.

CARTER, G. A. (1994): Ratios of leaf reflectances in narrow wavebands as indicators of plant stress. *Remote sensing*, 15(3), 697-703.

CURTIS, B.C., RAJARAM, S. AND GÓMEZ MACPHERSON, H. (2002): Bread Wheat; Improvement and Production. FAO Plant Production and Protection Series No. 30. FAO, Rome.

GITELSON, A. A., KAUFMAN, Y. J., STARK, R., AND RUNDQUIST, D. (2002): Novel algorithms for remote estimation of vegetation fraction. *Remote sensing of Environment*, 80(1), 76-87.

JELÍNEK, Z., STARÝ, K., KUMHÁLOVÁ, J., LUKÁŠ, J., AND MAŠEK, J. (2020): Winter wheat, winter rape and poppy crop growth evaluation with the help of remote and proximal sensing measurements.

LAZAREVIĆ, B., ŠATOVIĆ, Z., NIMAC, A., VIDAČ, M., GUNJAČA, J., POLITEO, O., AND CAROVIĆ-STANKO, K. (2021): Application of phenotyping methods in detection of drought and salinity stress in Basil (*Ocimum basilicum* L.). *Frontiers in Plant Science*, 12, 174.

LEŠIĆ, V., NOVAK, H., RATKOVIĆ, M., ZOVKO, M., LEMIĆ, D., SKENDŽIĆ, S., TABAK, J., POLIĆ, M., ORSAG, M. (2021): Rapid Plant Development Modelling System for Predictive Agriculture based on Artificial Intelligence // *Proceedings of the 16th International Conference on Telecommunications*, Zagreb, 2021. str. 173-180.

LILLESAND, T., KIEFER, R. W., AND CHIPMAN, J. (2014): *Remote Sensing and Image Interpretation*. New York, NY: John Wiley and Sons.

MAGNEY, T. S., EITEL, J. U., HUGGINS, D. R., AND VIERLING, L. A. (2016): Proximal NDVI derived phenology improves in-season predictions of wheat quantity and quality. *Agricultural and Forest Meteorology*, 217, 46-60.

RAHMAN, M. M., CRAIN, J., HAGHIGHATTALAB, A., SINGH, R. P., AND POLAND, J. (2021): Improving wheat yield prediction using secondary traits and high-density phenotyping under heat-stressed environments. *Frontiers in plant science*, 1977.

SEKAR, I., AND PAL, S. (2012): Rice and wheat crop productivity in the Indo-Gangetic Plains of India: changing pattern of growth and future strategies. *Indian Journal of Agricultural Economics*, 67(902-2016-67295).

XUE, J., AND SU, B. (2017): Significant remote sensing vegetation indices: A review of developments and applications. *Journal of sensors*, 2017.

WARRICK, R. A. (1988): Carbon dioxide, climatic change and agriculture. *Geographical journal*, 221-233.

THE EFFECT OF CHLORINE ON THE BIOFILM FORMATION OF *LISTERIA MONOCYTOGENES* STRAINS

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Summary: Food-borne diseases present a major problem throughout the world causing thousands of deaths each year from the consumption of food and water contaminated with toxin and pathogens. The most known organism linked with these diseases includes *Listeria monocytogenes*. Our aim is to study the effect of chlorine on different *Listeria* strains and their associated biofilms. This study evaluated the efficacy of chlorine treatment on the growth of *Listeria*. Therefore, nine strains were examined. We can notice that all *L. monocytogenes* strains formed decreasing biofilm amounts when they are treated with chlorine. However some strong and even weak biofilm forming strains showed considerable resistance to chlorine.

Keywords: *Listeria monocytogenes*, chlorine, biofilm formation

INTRODUCTION

Listeria monocytogenes has become an increasing cause for concern in the fresh produce industry. It is one of the main microorganisms that cause foodborne diseases and is able to grow at low temperatures as well (1 to 45 °C) (Lianou et al., 2017).

Listeria species are widespread in the environment, including soil, raw foods, stream water, plants, animals, and it also commonly colonize the food processing environment and ready to eat products, and contaminate deli meats, seafood, and dairy products. In 2016, the European Food Safety Authority described more than 2500 cases of listeriosis, with 247 deaths (EFSA, 2017).

Due to their widely distribution, *Listeria* species are used as hygiene indicators in all the stages of food processing chain. Biofilm formation allows bacteria to survive in unfavourable conditions and be persistent in food processing environments. In case when biofilms contain pathogenic bacteria it could cause contamination of food products and food-borne outbreaks.

To control the biofilm formation several methods are used such as chemical cleaning and sanitation. Many disinfecting or sanitizing agents often used in the food industry, such as quaternary ammonium compounds (QACs) and chlorine, have been shown to be effective against *L. monocytogenes* cells (Aarnisalo et al., 2000).

Therefore, our aim is to study the effect of chlorine on different *Listeria monocytogenes* strains and their associated biofilms.

The process of formation of a bacterial biofilm can be considered as a process that contain five developmental stages, as it can be seen in Figure 1.

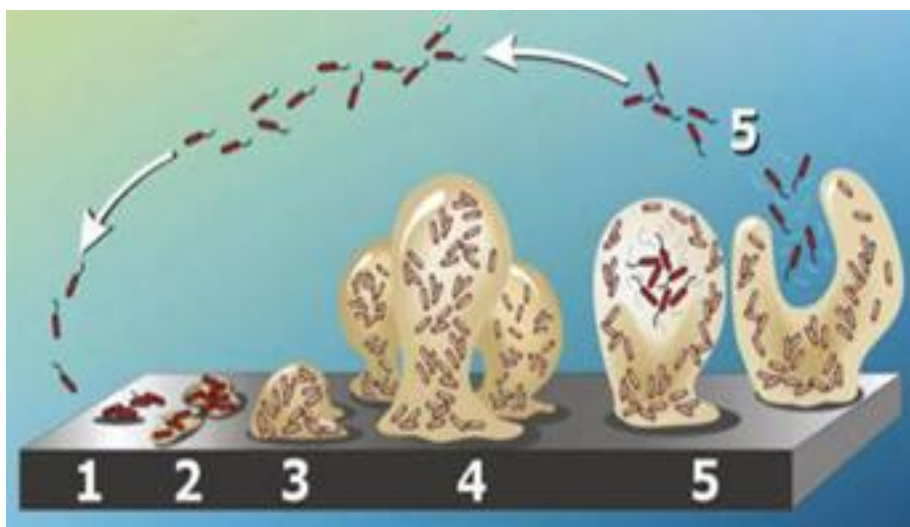


Figure 1. Process of biofilm formation: 1- Attachments of cells, 2- Cells proliferate and extrapolymeric substances (EPSs) are formed, 3- A three dimensional structure forms, 4- Maturation and 5-Deattachment of cells with recolonization (Van Houdt & Michielis, 2005)

MATERIAL AND METHODS

Nine *Listeria monocytogenes* isolates (NCAIMB1454, NCTC3b, 11/4.12t₀₃, CCM5576, CCM7202, NCAIM B01966^T, 10887^{1/2a}, 5105 3a, CCM 4699) were selected for this study. These isolates were obtained in a previous study in the Department of Food Microbiology, Hygiene and Safety of the Hungarian University of Agriculture and Life Sciences. The protocol of Biofilm formation used in this study was a modified version of the method of (Laura et al., 2018).

Fresh cultures were prepared in Trypto-Casein Soy Broth (TSB), cultivated at 37 °C for 24 hours. Biofilm formation was investigated using the crystal violet 0.4% (w/v) staining method (Musken et al., 2010).

Laboratory-generated biofilms were grown in an ELISA titer plate with a final volume of 200 µL of M9 minimal media (containing 1 g/L NH₄Cl, 6 g/L Na₂HPO₄, 3 g/L KH₂PO₄, 0.5 g/L NaCl, 2 ml/L MgSO₄ 1M, 0.1 ml/L CaCl₂ 1M, and 10 ml/L glucose 20%). Initial cell-counts were adjusted to an optical density of 0.3 using a McFarland densitometer. In order to study the behavior of *Listeria monocytogenes* strains when submitted to stress condition such as chemical cleaning, 200 µL of chlorine was added to each well of the ELISA titer plate.

Finally, in the last step, the optical density was measured using a microtiter plate reader (Multiscan Ascent, ThermoLab System) at 595 nm (OD₅₉₅). Multiscan Ascent instrument was controlled through a computer using the Ascent Software Version 2.6 (ThermoLabsystems). The measurement of the optical density (OD₅₉₅) of the microtiter plate wells was conducted using the method for biofilm read under specified parameters, by Multiscan Ascent instrument.

The measurement was conducted within 1-2 mins and the results were generated right after the measurement and could be imported to a statistical program to further conduct the statistical analysis of the obtained results. The OD595 results were imported to Microsoft Office Excel program. The average values and standard deviations of the optical density at 595 nm from the three biofilm assays were calculated for each *Listeria* strains. The standard deviations of the means were represented by error bars in the graphs.

RESULTS AND DISCUSSION

Biofilm formation of nine different *Listeria monocytogenes* strains was analyzed in M9 Minimal Media with and without chlorine, which results are shown in Figure 2.

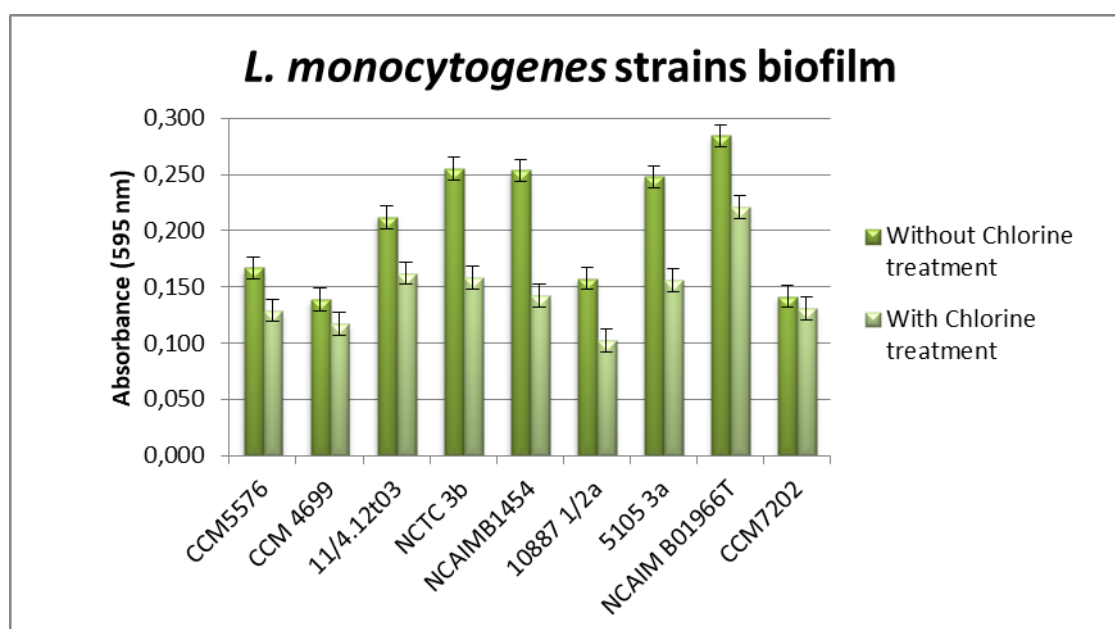


Figure 2. Biofilm formation of *L. monocytogenes* strains

Without chlorine treatment, *L. monocytogenes* NCAIM B01966T formed the highest amounts of biofilm (OD595 > 0.250). All the other strains showed similar biofilm amounts with an OD595 range from 0.150 to 0.250, while the weakest biofilms-forming strains found to be *L. monocytogenes* CCM4699 and CCM7202 showing an OD595 < 0.150 under the same conditions.

We can notice that all *L. monocytogenes* strains formed biofilm in a lesser extent when chlorine is added. It is interesting, that cleaning agents that are mainly designed to remove organic matter eliminated *L. monocytogenes*. In a previous study of Aarnisalo et al. (2000) a chlorinated alkaline cleaner was also found to have good disinfecting properties against *L. monocytogenes*.

The biggest decrease of biofilm formation was observed in case of *L. monocytogenes* NCAIMB1454, 5105 3a and NCTC3b strains, where the OD595 which was higher than 0.250 dropped to ± 0.150 with chlorine treatment.

L. monocytogenes NCAIM B01966T strain found to be the biggest biofilm forming strain showing resistance to chlorine. Although strains CCM4699 and CCM7202 proved to be weak biofilm formers compared to other strains they also showed considerable resistance compared to their non- treated counterparts. Whereas *L. monocytogenes* 10887 1/2a showed the lowest biofilm amount.

CONCLUSION

As our preliminary studies showed, the chlorine treatment accentuates the inhibition of *L. monocytogenes*, and it decreased their associated biofilms however some strong and even weak biofilm forming strains showed considerable resistance to chlorine. A detailed study on the effect of other disinfecting or sanitizing agents on biofilm formation of different *Listeria monocytogenes* strains is needed.

*

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REFERENCES

- AARNISALO, K., SALO, S., MIETTINEN, H., SUIHKO, M.-L., WIRTANEN, G., . AUTIO, T. (2000). Bactericidal efficiencies of commercial disinfectants against *Listeria monocytogenes* on surfaces. *Journal of Food Safety*, 20(4), 237–250.
- EFSA - EUROPEAN FOOD SAFETY AUTHORITY (2017). The European Union summary report on trends and sources of zoonoses, zoonotic agents and food-borne outbreaks in 2016. <https://www.efsa.europa.eu/en/efsajournal/pub/5077>.
- LIANOU, A., PANAGOU, E. Z., & NYCHAS, G.-J. E. (2017). Meat safety—I foodborne pathogens and other biological issues. In F. Toldrá (Ed.). *Lawrie's Meat Science* (pp. 521–552). (8th ed.). Springer. <https://doi.org/10.1016/B978-0-08-100694-8.00017-0>.
- LAURA LUQUE-SASTRE, EDWARD M. FOX, KIERAN JORDAN, SÉAMUS FANNING. 2018. A Comparative Study of the Susceptibility of *Listeria* Species to Sanitizer Treatments When Grown under Planktonic and Biofilm Conditions.
- MUSKEN, M., S. DI FIORE, U. ROMLING, AND S. HAUSSLER. 2010. A 96-well-plate–based optical method for the quantitative and qualitative evaluation of *Pseudomonas*

aeruginosa biofilm formation and its application to susceptibility testing. Nat. Protoc. 5:1460–1469.

VAN HOUDT, R. AND MICHIELS, C.W., 2005. Role of bacterial cell surface structures in Escherichia coli biofilm formation. *Research in microbiology*, 156(5-6), pp.626-633

MICROENCAPSULATION USING SPRAY DRYING TO PROTECT OLIVE OIL AGAINST OXIDATION – A REVIEW

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Summary: *Virgin olive oil (VOO) is widely produced and consumed in the mediterranean region. It is highly appreciated for its nutritional properties due to its high content in volatile and non-volatile compounds of interest. In addition, olive oil presents considerable amounts of natural antioxidants and is related with many health benefits, including the prevention of many modern life-style diseases, like some kinds of cancer and cardiovascular diseases. Nevertheless, it is chemically unstable and susceptible to degradation. Microencapsulation is an effective and important tool to prepare oil-based high-quality and health-beneficial products in various industries in order to enhance their chemical, oxidative, and thermal stability. Emulsification technology is a key step in the microencapsulation of oils. Thus, one of the solutions to preserve olive oil's bioactives is microencapsulation to get it as a powder form. The purpose of this article is to provide a review on microencapsulation of olive oil by spray drying using the appropriate wall materials. We first recall the main aspects of oil deterioration. Then, we present the wall material selection for the step of membrane emulsification, and finally we present the main studies on the preparation of encapsulated olive oils.*

Key words: *Oxidation, olive oil, bioactives, wall materials, microencapsulation.*

INTRODUCTION

Virgin olive oil (VOO) is obtained from the fruit of the olive tree (*Olea europea L*) and is widely produced and consumed in the Mediterranean region. It is highly appreciated for its fine taste and aroma, as well as for its nutritional properties due to its high content in volatile and non-volatile compounds of interest. In addition, olive oil presents considerable amounts of natural antioxidants (Moldão-Martins et al., 2004). The nutritional benefits of VOO are primarily related to the fatty acid composition, mainly due to the high content of oleic acid and also to the balanced ratio of saturated and polyunsaturated fatty acids that are known to show protective effects against cardiovascular diseases by preventing oxidation of low density lipoproteins. Besides, the content of phenolic compounds is an important factor to be considered in evaluating VOO quality. In fact, phenols are powerful antioxidants and a significant contributor to its stability against oxidants, and hence to extending its shelf-life relative to other vegetable oils (Veillet, Tomao, & Chemat, 2010). Moreover, olive oil is

related with many health benefits, including the prevention of many modern lifestyle diseases, like some kinds of cancer and cardiovascular diseases (Sousa et al., 2015).

Therefore, because of its high instauration degree, olive oil is subject to oxidation during processing, distribution and handling. Oxidation leads to the formation of unpleasant tastes and odors and, consequently, to the reduction of oil's shelf life, besides promoting the generation of free radical, which may have negative physiological effects on the organism. Thus, one of the solutions to preserve these bioactives is the microencapsulation:

Microencapsulation is a method in which tiny particles or droplets are surrounded by a coating wall, or are embedded in a homogeneous or heterogeneous matrix, to form small capsules (Rodríguez, Martín, Ruiz, & Clares, 2016). The key step to release capsules is the emulsification technology (Bakry et al., 2016). It is generally applied for the encapsulation of bioactives in aqueous solutions.

Basically, an emulsion consists of at least 2 immiscible liquids, usually oil and water, with one of the liquids being dispersed as small spherical droplets in the other. Dispersion by artificial membranes, or as widely known Membrane Emulsification, is a novel unit operation for producing single and multiple emulsions or complex suspension/emulsion fluids. The dispersed phase is forced through the pores of microporous membrane directly into the continuous phase by using low pressure. Emulsified droplets are formed and detached at the end of the pores with drop by drop mechanism (Joscelyne & Trägårdh, 2000). The purpose of this article is to provide a review on microencapsulation of olive oil by spray drying using the appropriate wall materials. We first recall the main aspects of oil deterioration. Then, we present the wall material selection for the step of membrane emulsification, and finally we present the main studies on the preparation of encapsulated olive oils

HYDROLYTIC AND OXIDATIVE RANCIDITY PHENOMENA IN OLIVE OIL

The main processes leading to the deterioration of lipids are hydrolytic rancidity (lipolysis) and oxidative rancidity (oxidation). In olive oil, the former usually begins while the oil is in the fruit, whereas the latter mainly occurs during the extraction process and storage. Oxidation can occur either in the dark (autoxidation) or in the presence of light (photo-oxidation) (Koidis & Boskou, 2014).

1. Hydrolysis

Lipolysis or hydrolysis is caused by the release of free fatty acids (FFAs) from glycerides. Hydrolytic rancidity is extremely important in determining how a product tastes; it is unlikely to be of any nutritional significance because fats are in any case enzymatically hydrolyzed in the small intestine before they are absorbed. Olive oil lipolysis results from microbial and enzymatic lipolysis. Olive oil lipolysis is the most important cause of negative changes and happens by enzymatic, microbial and chemical reaction pathways (Ramon Aparicio & Harwood, 2013).

1.1. Enzymatic pathway

The degradation of lipids in plant tissues is performed by a process often called the polyunsaturated fatty acid (PUFA) cascade. Typically, the decomposition of unsaturated fatty acids requires different enzymes as shown in Figure 1. The sequence begins with the

hydrolysis of various acylglycerides by lipases, lipolytic acyl hydrolases and phospholipases, and free PUFAs are released. Lipoxygenases then convert unsaturated fatty acids into two main hydroperoxides, namely 9 and 13 isomers, which are unstable. In the last step of the cascade, lyases, isomerases, and dehydrogenases degrade hydroperoxides into a variety of volatile and nonvolatile products. The flavor components formed, such as aldehydes, ketones, and alcohols, can be directly responsible for the off-flavor occurrence in oils (Aparicio & Harwood, 2013). In olives the formation of 13-hydroperoxides from linoleic and linolenic acids is promoted, leading to the formation of C6 aldehydes and alcohols, which are the main contributors to VOO sensory perception. Free unsaturated fatty acids, particularly linoleic and linolenic in plants, are the preferred substrates for oxidation by lipoxygenases.

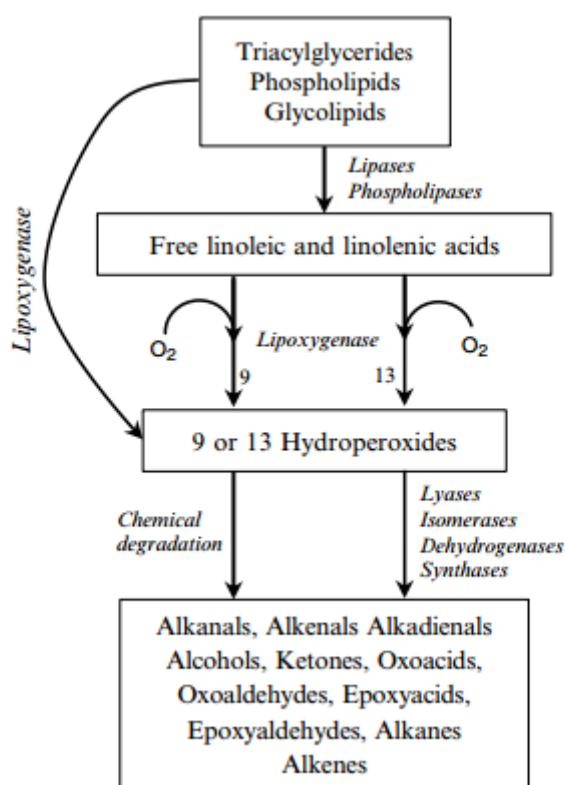


Fig. 1 Enzymatic oxidation of unsaturated fatty acids. Numbers with lipoxygenase represent enzymes producing specific isomers of hydroperoxide (Ramon Aparicio & Harwood, 2013).

1.2. Microbial pathway

Microbial enzymes cannot be excluded as an additional deterioration factor. The microorganisms present in the olive fruit liberate the enzyme lipase responsible for the microbial lipolysis.

The formation of small amounts of the hydroperoxides can have an accelerating effect on oxidation in the finished oil. Free radicals formed from the decomposition of hydroperoxides can escalate further oxidation, causing earlier-than-expected off-flavor formation and lowering oil storage stability (Morales and Przybylski, 2013).

2. Lipid oxidation

The reaction of oxygen with unsaturated fatty acids (LH) involves a free radical mechanism of lipid oxidation which is usually divided into three stages: initiation, propagation, and termination. The oxidation of lipids is accelerated by many factors such as light, temperature, enzymes, metals, metalloproteins, pigments, and microorganisms.

In the initiation stage free radicals are formed directly from fatty acids in the presence of initiators such as temperature, light, other free radicals, and heavy metals. The resulting lipid free radicals ($L\cdot$) react with oxygen to form peroxy radicals ($LOO\cdot$). Formed at this propagation stage, peroxy radicals react with another molecule of lipid (LH), forming a lipid radical and a hydroperoxide ($LOOH$) which is a fundamental primary product of auto-oxidation. During the termination stage, radicals react with each other and form non-radical products. Any reaction that prevents the propagation of peroxidation or removes free radicals from the system plays a key role in the termination mechanism (Frankel 1984; Morales and Przybylski, 2013):

Initiation $LH \rightarrow L\cdot + H\cdot$

Propagation $L\cdot + O_2 \rightarrow LOO\cdot$

$LOO\cdot + LH \rightarrow LOOH + L\cdot$

Termination $L\cdot + L\cdot \rightarrow LL$

$L\cdot + LOO\cdot \rightarrow LOOL$

$LOO\cdot + LOO\cdot \rightarrow LOOL + O_2$

Metallic ion catalysts in the presence of small amounts of hydroperoxides are the most important initiators of lipid oxidation. Transition metals such as iron and copper catalyze both the initiation and decomposition of hydroperoxides.

3. Photosensitized oxidation

Photosensitized oxidation has been recognized as the most detrimental factor to the oxidative stability of vegetable oils. Exposure to light can cause the formation of hydroperoxides when both oxygen and the photosensitizer (chlorophylls and pheophytins) are present. The mechanism of photosensitized oxidation consists of the transfer of energy ($h\nu$) from light onto the sensitizer (Sensitizer Excited), which in turn can react directly with a lipid (RH), forming radicals (R^*) that in turn initiate autoxidation. More damaging is the reaction of an excited sensitizer with ground state oxygen to form singlet oxygen. The transfer of energy onto an acceptor, an unsaturated fatty acid, causes the formation of a hydroperoxide. Singlet oxygen has been found to react with linoleic acid about 1,500 times faster than normal oxygen. Therefore, this reactive component is defined as the most important initiator in the free radical autoxidation of vegetable oils. The effectiveness of light initiation is inversely proportional to the wavelength, where shorter wavelengths are the most effective. This process is propagated by the free radical mechanism discussed earlier, and similar hydroperoxides are formed (Morales and Przybylski, 2013).

To protect the oil from all of the deterioration mechanisms listed above, we will proceed to encapsulate olive oil to preserve it. Thus, emulsifying technology is a key step to obtain the dried form of olive oil.

MICROENCAPSULATION BY SPRAY DRYING OF OLIVE OIL

Emulsification is an important step to control since it conditions the droplet size distribution, viscosity and stability of the emulsion. Oil encapsulation efficiency is higher when emulsions are monodisperse and droplet size is smaller. Other quality parameters of microparticles such as morphology, oxidation stability and surface-free oil content define the stability and the composition of the emulsion (Ramakrishnan & Ferrando, 2014). Emulsions are liquid systems composed of two immiscible phases referred to as continuous and disperse phase, respectively. In O/W emulsions, the oil is the dispersed phase and water is the continuous phase. Alternatively, in water-in-oil emulsions (W/O), the water is dispersed phase and the continuous phase is oil. Multiple emulsions can also be produced, like water-in-oil-in-water (W/O/W) or oil in-water-in-oil (O/W/O) emulsions (Dickinson, 2009; McClements, 2018). Microcapsules or microspheres are widely used in many different applications in chemical, food, cosmetic and pharmaceutical industries to encapsulate protect and deliver bioactive compounds.

Microencapsulation is used in food industry to reduce the core reactivity with environmental factors such as heat, air and moisture; to decrease the transfer rate of the core material to the outside environment; to promote easier handling; to control the release of the core material; to mask the core taste; to dilute the core material when it should be used in only very small amounts; to separate components within a mixture that would otherwise react with one another (Desai & Jin Park, 2005; Gharsallaoui, Roudaut, Chambin, Voilley, & Saurel, 2007). The microencapsulation process steps are first the choice of the wall material, second the emulsification technology and third the spray drying technology.

1. Selection of wall material for olive oil microencapsulation

The initial step in microencapsulation of the food ingredient is to make the choice of a suitable coating material, basically a film-forming biopolymer, from a wide variety of natural or synthetic polymers, depending on the core material and the characteristics desired in the final microcapsules (Jafari, Assadpoor, He, & Bhandari, 2008).

Selecting the most appropriate wall material is an important step in microcapsules development (Calvo, Hernández, Lozano, & González-Gómez, 2010). In fact, the composition and structure of the established wall influences the efficiency of protection or controlled release (Calvo, Castaño, Lozano, & González-Gómez, 2012).

Wall material could act as a barrier and it may protect against oxygen, water, and light or could avoid contact with other ingredients or control diffusion. Wall material for microencapsulation of oil must have high stability, high water solubility, drying properties, a tendency to form a fine and dense network during drying, emulsifying properties and should not permit lipid separation from the emulsion during dehydration, for spray-drying they must have low viscosity too (Calvo et al. 2012; Gharsallaoui et al. 2007).

Because almost no wall material can meet all the properties listed, in practice they are used in combination with each other (Hogan, McNamee, O'Riordan, & O'Sullivan, 2001; Jafari et al., 2008).

Concerning microencapsulation by spraydrying, maltodextrin, gum arabic and modified starch are the most widely used because those polymers are of high molecular weight and high glass transition temperature (T_g) (Samantha et al., 2015).

Wall materials could be made by starches like maltodextrin and modified starches, proteins such as milk proteins and gelatine, lipids (tristearin, stearic acid, cholesterol, cetylpalmitate), resins (Shellac) and synthetic polymers derived from esters of acrylic and methacrylic acid, gums like acacia gum and mesquite gum and other materials like cyclodextrins and liposomes (Jafari et al., 2008; Martins et al., 2017).

There common properties are summarized into five: First no reactivity with the core material. Second, their ability to provide maximum protection during the process of emulsification and drying and storage against environmental conditions. Third, good economy and functionality. Fourth, complete release of the core material under desirable conditions and finally good rheological properties and solubility (Jafari et al. 2008).

Starches

Hydrolysed starch and maltodextrins in combination with other wall materials with emulsifying capacity are commonly used to encapsulate hydrophobic food ingredients. The advantages of using starch hydrolysates include low cost, high water solubility, and bland flavour (Zhu, 2017). Their application for encapsulation of olive oil has been reviewed recently. A recent example was using starch as wall material and it results in maintaining the oxidative stability of the oil comparing with free oil (Lei et al., 2018).

Maltodextrins

Maltodextrins $[(C_6H_{12}O_5)_nH_2O]$ are saccharides non-sweet polymers, which consist of D-glucose unities joined by α 1-4 chain. Sold as white powders, maltodextrins are formed by partially hydrolysing corn starch with acids and/ or enzymes. They consist of glucose polymers of various lengths. The molecular weight of 10 DE (dextrose equivalent) is approximately 1800 daltons. DE indicates the level of hydrolysis of the starch polymer. Thus, the DO of a product of hydrolysis is equivalent to its reducing power as a percentage of the reducing power of the pure dextrose (D-glucose), therefore inversely proportional to its average molecular weight (Gharsallaoui et al., 2007).

Viscosities of maltodextrins are lower than gum arabic and they have no lipophilic groups. That's why their emulsification properties are poor. Their advantages include low relative cost (one third of modified starch) low flavour, used at high solids concentrations and improvement of the shelf-life of oils by protecting them against oxidation (Yakdhane et al., 2021).

Gum arabic

Gum arabic is a polymer consisting of d-glucuronic acid, l-rhamnose, d-galactose and l-arabinose, including approximately 2% protein.

It is preferred instead of other types of gums because of its excellent emulsifying capacity. It is effective at low pH, at high ionic strength, and in the presence of beverage colouring agents. However, its use is difficult due to its high cost and availability problems because a rather high gum/oil ratio (w1:1) is required in order to produce fine stable emulsion droplets ($d_{32} \ll 1$ mm) (Dickinson, 2009). Moreover, it is only produced in regions subjected to unpredictable climatic variations. Therefore other materials are being investigated as

alternatives such as maltodextrins and modified starches which were used to partially replace it (Liu & Yang, 2011).

The use of gum arabic as an encapsulating matrix is common due to its characteristics of viscosity, solubility and emulsification. It was traditionally used for lipid encapsulation and can produce stable emulsions with most oils (Gharsallaoui et al. 2007).

Modified starch

Starch is a widely available and low cost product. Besides, the viscosity of starch solutions is generally too high which is good for the drying process. Starch does not have hydrophobic groups, therefore does not exert any stabilizing effect in the emulsions, except for an increase in viscosity. The production of modified starch is an alternative that has been developed for some time, aiming to overcome a few limitations of the native starches, thereby increasing use and applicability of this polymer in the industry. The reasons that lead to the modification of starch include: modifying the characteristics of gelatinization; decreasing the downgrading and the tendency of pastes to turn into gels; increasing stability of pastes in cooling and defrosting; the transparency of pastes and gels and their adhesive qualities; improving the texture of pastes and gels and the formation of films; adding hydrophobic groups and introducing emulsifying power (Silva, Morelli, Dragosavac, Starov, & Holdich, 2017).

Milk proteins

Whey proteins have been reported to be an effective wall material in oil encapsulation with spray drying such as linseed oil (Gallardo et al., 2013) due to the formation of thick viscoelastic films at droplets interfaces of emulsion thus inhibition the lipid oxidation (Koç et al., 2015).

Regarding olive oil microencapsulation, very little information is available in the literature. The wall materials combinations used to encapsulate olive oil are summarized in table1. Calvo, Hernández, Lozano, & González-Gómez, 2010, used different wall components such as proteins (sodium caseinate and gelatin), hydrocolloids (Arabic gum), and hydrolyzed starches (starch, lactose, and maltodextrin). The highest microencapsulation yield was observed when gelatine, maltodextrin and Arabic gum were used as wall materials.

Sun-Waterhouse, Zhou, Miskelly, Wibisono, & Wadhwa, 2011, also investigated the influence of microencapsulation and addition of a phenolic compound (caffeic acid) on the storage stability of olive oils encapsulated in alginate shell at a level of 50 mg oil/g alginate solution.

In 2012, calvo et al. showed that maltodextrin, carboxymethylcellulose and lecithin when used together as encapsulating agents gave the highest microencapsulation yields.

Devi, Hazarika, Deka, & Kakati, 2012, showed that olive oil could be encapsulated efficiently using gelatin A-sodium alginate complex using glutaraldehyde as crosslinker.

Koç et al., 2015, targeted the optimization of process variables to obtain extra virgin olive oil powder to have maximum microencapsulation efficiency and minimum particle size. Thus, wall material composition using a mixture of maltodextrin and whey protein isolate, homogenization rate, and feeding rate for spray dryer were optimized to find that high microencapsulation efficiency of extra virgin olive oil is achieved when the wall materials composition is 92 % maltodextrin, 7 % whey protein isolate, 1 % Tween 20, and 17,500 rpm of rotor-stator homogenization rate and 22 % of pump rate, respectively.

2. Encapsulation process by Spray drying

Many encapsulation processes are used such as spray-drying, spray-cooling, spray-chilling, air suspension coating, extrusion, centrifugal extrusion, freeze-drying, coacervation, rotational suspension separation, co-crystallization, liposome entrapment, interfacial polymerization and molecular inclusion (Gharsallaoui et al., 2007). In our research, we will be focusing on spray drying method.

Microencapsulation by spray-drying has been successfully used in the food industry for several decades, and this process is one of the oldest encapsulation methods used since the 1930s to prepare the first encapsulated flavors using gum acacia as wall material (Calvo et al. 2010).

Various oils in the food industry where encapsulated successfully using this technique. Recently, several studies have been undertaken to encapsulate the oils by spray-drying in food (Bakry et al., 2016; Carneiro, Tonon, Grosso, & Hubinger, 2013; Le et al., 2017; Rubilar et al., 2012; Victória, Fernandes, & Vilela, 2014; Zhu, 2017) cosmetics (Carvalho, Silva, & Hubinger, 2014), pharmaceutical (Liu & Yang, 2011), and pesticide industries (López et al., 2014).

Food industry refers to spray drying in order to ensure a microbiological stability of products, avoid the risk of chemical and/or biological degradations, reduce the storage and transport costs, and finally obtain a product with specific properties like instantaneous solubility for example. It is also used due to its low cost and available equipment.

The application of spray-drying process in microencapsulation involves three basic steps:

- Preparation of the dispersion or emulsion to be processed;
- homogenization of the dispersion;
- Atomization of the mass into the drying chamber.

The mechanism of this technique includes the homogenization of the material for encapsulation with the carrier material at a different ratio. The mixture is then fed into a spray dryer and atomized with a nozzle or spinning wheel. Water is evaporated by the hot air contacting the atomized material. The microcapsules are then collected after they fall to the bottom of the drier. One limitation of the spray-drying technology is the limited number of wall materials available which are expected to have good water solubility. Typical wall materials include gum acacia, maltodextrins, hydrophobically modified starch, and mixtures of them. Other polysaccharides (alginate, carboxymethylcellulose, guar gum) and proteins (whey proteins, soy proteins, sodium caseinate) can be used as the wall material in spray-drying, but their usage becomes very tedious and expensive because of their low solubility in water. Another limitation is that spray drying provides fine microcapsules powder which needs further processing such as agglomeration. In addition, Encapsulated materials have lower oxidative stability due to the high temperatures used during the atomization process (Bakry et al., 2016; Gharsallaoui et al., 2007).

CONCLUSION

Olive oil has promising potentials for maintaining and promoting health, as well as preventing and potentially treating some diseases. However, its low stability has limited its application as food additive, nutritional supplement, and active cosmetic or as drugs in different industrial

areas. Microencapsulation is a well-established methodology for the preservation of olive oil bioactive ingredients. It offers many advantages, including effective protection of the encapsulated oil against degradation, the possibility of accurately control release, easy administration and avoiding the evaporation of volatile compounds. Various encapsulating agents could be used for olive oil encapsulation however the most common are the carbohydrates, due to showing a higher affinity with various types of materials to be encapsulated.

REFERENCES

- APARICIO, R., & HARWOOD, J. (2013). Handbook of olive oil: Analysis and properties. Handbook of Olive Oil: Analysis and Properties. <https://doi.org/10.1007/978-1-4614-7777-8>
- BAKRY, A. M., ABBAS, S., ALI, B., MAJEED, H., ABOUELWABA, M. Y., MOUSA, A., & LIANG, L. (2016). Microencapsulation of Oils: A Comprehensive Review of Benefits, Techniques, and Applications. *Comprehensive Reviews in Food Science and Food Safety*, 15(1), 143–182.
- CALVO, P., CASTAÑO, Á. L., LOZANO, M., & GONZÁLEZ-GÓMEZ, D. (2012). Influence of the microencapsulation on the quality parameters and shelf-life of extra-virgin olive oil encapsulated in the presence of BHT and different capsule wall components. *Food Research International*, 45(1), 256–261.
- CALVO, P., HERNÁNDEZ, T., LOZANO, M., & GONZÁLEZ-GÓMEZ, D. (2010). Microencapsulation of extra-virgin olive oil by spray-drying: Influence of wall material and olive quality. *European Journal of Lipid Science and Technology*, 112(8), 852–858.
- CARNEIRO, H. C. F., TONON, R. V, GROSSO, C. R. F., & HUBINGER, M. D. (2013). Encapsulation efficiency and oxidative stability of flaxseed oil microencapsulated by spray drying using different combinations of wall materials. *Journal of Food Engineering*, 115(4), 443–451.
- CARVALHO, A. G. S., SILVA, V. M., & HUBINGER, M. D. (2014). Microencapsulation by spray drying of emulsified green coffee oil with two-layered membranes. *Food Research International*, 61, 236–245.
- DESAI, K. G. H., & JIN PARK, H. (2005). Recent Developments in Microencapsulation of Food Ingredients. *Drying Technology*.
- DICKINSON, E. (2009). Hydrocolloids as emulsifiers and emulsion stabilizers. *Food Hydrocolloids*, 23(6), 1473–1482.
- FRANKEL, E. N. (1984). Lipid oxidation: Mechanisms, products and biological significance. *Journal of the American Oil Chemists' Society*.
- GALLARDO, G., GUIDA, L., MARTINEZ, V., LÓPEZ, M. C., BERNHARDT, D., BLASCO, R., HERMIDA, L. G. (2013). Microencapsulation of linseed oil by spray drying for functional food application. *FRIN*, 52(2), 473–482.
- GHARSALLAOUI, A., ROUDAUT, G., CHAMBIN, O., VOILLEY, A., & SAUREL, R. (2007). Applications of spray-drying in microencapsulation of food ingredients: An overview. *Food Research International*.

- HOGAN, S. A., MCNAMEE, B. F., O'RIORDAN, E. D., & O'SULLIVAN, M. (2001). Emulsification and microencapsulation properties of sodium caseinate/carbohydrate blends. *International Dairy Journal*, 11(3), 137–144.
- JAFARI, S. M., ASSADPOOR, E., HE, Y., & BHANDARI, B. (2008). Encapsulation Efficiency of Food Flavours and Oils during Spray Drying Spray Drying, 3937.
- JOSCELYNE, S. M., & TRÄGÅRDH, G. (2000). Membrane emulsification-a literature review. *Journal of Membrane Science* (Vol. 169).
- KOÇ, M., GÜNGÖR, Ö., ZUNGUR, A., YALÇIN, B., SELEK, İ., ERTEKIN, F. K., & ÖTLES, S. (2015). Microencapsulation of Extra Virgin Olive Oil by Spray Drying: Effect of Wall Materials Composition, Process Conditions, and Emulsification Method. *Food and Bioprocess Technology*, 8(2), 301–318.
- KOUIDIS, A., & BOSKOU, D. (2014). Virgin Olive Oil: Losses of Antioxidant Polar Phenolic Compounds due to Storage, Packaging, and Culinary Uses. Processing and Impact on Active Components in Food. Elsevier Inc.
- LE, T. H., TRAN, T. M. V., TON, N. M. N., TRAN, T. T. T., HUYNH, T. V., NGUYEN, T. N., LE, V. V. M. (2017). Combination of whey protein and carbohydrate for microencapsulation of pumpkin (*Cucurbita* spp.) seed oil by spray-drying. *International Food Research Journal*, 24(3), 1227–1232.
- LEI, M., JIANG, F. C., CAI, J., HU, S., ZHOU, R., LIU, G., XIONG, X. G. (2018). Facile microencapsulation of olive oil in porous starch granules: Fabrication, characterization, and oxidative stability. *International Journal of Biological Macromolecules*, 111, 755–761.
- LIU, T. T., & YANG, T. S. (2011). Optimization of Emulsification and Microencapsulation of Evening Primrose Oil and Its Oxidative Stability During Storage By Response Surface Methodology. *Journal of Food Quality*, 34(1), 64–73.
- LÓPEZ, A., CASTRO, S., ANDINA, M. J., URES, X., MUNGUÍA, B., LLABOT, J. M., DOMÍNGUEZ, L. (2014). Insecticidal activity of microencapsulated *Schinus molle* essential oil. *Industrial Crops and Products*, 53, 209–216.
- MARTINS, E., PONCELET, D., RODRIGUES, R. C., MARTINS, E., PONCELET, D., RODRIGUES, R. C., & RENARD, D. (2017). Oil encapsulation techniques using alginate as encapsulating agent : applications and drawbacks. *Journal of Microencapsulation*, 0(0), 754–771.
- MCCLEMENTS, D. J. (2018). Advances in nanoparticle and microparticle delivery systems for increasing the dispersibility, stability, and bioactivity of phytochemicals. *Biotechnology Advances*, (August), 0–1.
- MOLDÃO-MARTINS, M., BEIRÃO-DA-COSTA, S., NEVES, C., CAVALEIRO, C., SALGUEIRO, L., & LUÍSA BEIRÃO-DA-COSTA, M. (2004). Olive oil flavoured by the essential oils of *Mentha × piperita* and *Thymus mastichina* L. *Food Quality and Preference*, 15(5), 447–452.
- MORALES, M. T., & PRZYBYLSKI, R. (2013). Handbook of olive oil. Analysis and Properties, Springer Science+Business Media New York, 480-515.
- RAMAKRISHNAN, S., & FERRANDO, M. (2014). Influence of Emulsification Technique and Wall Composition on Physicochemical Properties and Oxidative Stability of Fish Oil Microcapsules Produced by Spray Drying, 1959–1972.

RODRÍGUEZ, J., MARTÍN, M. J., RUIZ, M. A., & CLARES, B. (2016, May 1). Current encapsulation strategies for bioactive oils: From alimentary to pharmaceutical perspectives. *Food Research International*. Elsevier Ltd.

RUBILAR, M., MORALES, E., CONTRERAS, K., CEBALLOS, C., ACEVEDO, F., VILLARROEL, M., & SHENE, C. (2012). Development of a soup powder enriched with microencapsulated linseed oil as a source of omega-3 fatty acids. *European Journal of Lipid Science and Technology*, 114(4), 423–433.

SAMANTHA, S. C., BRUNA, A. S. M., ADRIANA, R. M., FABIO, B., SANDRO, A. R., & ALINE, R. C. A. (2015). Drying by spray drying in the food industry: Micro-encapsulation, process parameters and main carriers used. *African Journal of Food Science*, 9(9), 462–470.

SILVA, P. S., MORELLI, S., DRAGOSAVAC, M. M., STAROV, V. M., & HOLDICH, R. G. (2017). Water in oil emulsions from hydrophobized metal membranes and characterization of dynamic interfacial tension in membrane emulsification. *Colloids and Surfaces A: Physicochemical and Engineering Aspects*.

SOUSA, A., CASAL, S., MALHEIRO, R., LAMAS, H., BENTO, A., & PEREIRA, J. A. (2015). Aromatized olive oils: Influence of flavouring in quality, composition, stability, antioxidants, and antiradical potential. *LWT - Food Science and Technology*, 60(1), 22–28.

VEILLET, S., TOMAO, V., & CHEMAT, F. (2010). Ultrasound assisted maceration: An original procedure for direct aromatisation of olive oil with basil. *Food Chemistry*, 123(3), 905–911.

VICTÓRIA, R., FERNANDES, D. B., & VILELA, S. (2014). Gum arabic / starch / maltodextrin / inulin as wall materials on the microencapsulation of rosemary essential oil. *Carbohydrate Polymers*, 101, 524–532.

YAKDHANE, A., LABIDI, S., CHAABANE, D., TOLNAY, A., NATH, A., KORIS, A., & VATAI, G. (2021). Microencapsulation of Flaxseed Oil—State of Art. *Processes*, 9(2), 295.

ZHU, F. (2017, August 15). Encapsulation and delivery of food ingredients using starch based systems. *Food Chemistry*. Elsevier Ltd.

THE USE OF REDOX POTENTIAL AS AN ALTERNATIVE FOR BACTERIAL RECOGNITION – A REVIEW

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Summary: *Biological systems change their redox potential due to the natural oxidation-reduction reactions. The energy source of the microbial growth is the biological oxidation which results in a reduction in the environment. This is because of the oxygen depletion and the production of reducing compounds in the nutrient medium. The typically decreasing redox potential shows specific kinetics depending on the microorganism and its metabolism. Obtaining quantitative indices allows to describe the unique shape of redox potential curves of different microorganisms or microorganism groups. The redox potential curve is recorded over a time range, and it helps to define reference points and stable parameters.*

Keywords: *Redox potential, Redox potential curves, microbial growth.*

INTRODUCTION

The determination of food microbiological quality is important to guarantee the food safety. Thus, microbiological analyses are integrated in order to control the hygiene of food chain and to detect emerging risks. Until now, the detection of microorganisms is mostly applied by the standard method, which is colony count (Feng, 1996). The principle of this method is the determination of the total number of bacteria in a specific product by inoculating a well-defined volume from dilutions of suspensions of the product onto a solid growth medium. During incubation, the cells multiply to form colonies visible by naked eye. Incubation time and temperature are fixed according to the specific growth conditions of the bacteria after incubation enumeration is preformed and the results are presented in cfu/ml for liquid or cfu/g (Jasson et al., 2010). However, this conventional method is a slow method as the incubation period is long and can even last up to 168 hours in some cases (Erdosi et al., 2012). Therefore, a great attention has been placed to develop faster alternative methods. Quick alternatives include impedimetric method and redox potential method. The impedimetric method is based on the detection of impedance change of a special broth that develops according to the proliferation of microorganisms. It is true that this method is less

time consuming than the colony counting method, but it is expensive (Reichart et al., 2007). In this context, authors elaborated redox potential measurement-based new method. It is a simple, rapid, and useful procedure for the examination of microbial activity. This method is based on the most complex factor of the microbiological condition which is the redox potential, this factor depends on the oxygen reduction and the production of reducing compounds in the environment (Szakmár et al., 2014).

RECOGNITION OF MICROORGANISMS BY DIFFERENT METHODS

Conventionally, the standard methods adopted for bacterial count using the cultivation and colony counting is the culturing method, and it is well established, but the cultivation process requires typically one to several days to complete the inspection which requires to develop rapid alternatives methods for bacterial quantification and qualification in a limit period. The rapid method must have a high sensibility and afford a rapid-detection and a correct diagnosis and for sure inexpensive. Impedance-measurement is a rapid alternative method for quantitative and qualitative detection of bacteria (Reichart et al., 2007).

The method is based on detection of the change in the impedance of a special broth, which occurs in function of microorganisms' proliferation. The impedance microbiology technique has developed into a well-elaborated rapid method for both qualitative and quantitative detection of different microorganisms. This method shows a decreasing linear relationship between the time required to note a significant increase in conductance which is called "Time To Detection" (TTD) and the logarithm of the initial level of microorganisms in the range of $10^3 - 10^7$ cells/ml and this relationship serve for calculating the original living cell concentration from the TTDs (Reichart et al., 2007).

This method has several weaknesses, it is an expensive method because it requires special nutrient broths with originally low impedance and a specific thermo-regulation, the detection period becomes longer in the case of lower bacterial concentration because in order to guarantee a significant electrical signal, it is necessary to have larger loads of the bacteria, so the logarithm of the original viable cell number is uncertain (R. Hamada et al, 2003). The selective media that contains high salt concentrations used in microbiology can result in a high conductance reading, so it will be classified outside the normal validated range of the impedimetric systems. the indirect technique is proposed to avoid this problem. This indirect method is based on monitoring microbial metabolism through CO₂ production. However, this method cannot be used for some bacteria as they do not produce CO₂. Thus, with the purpose of avoiding the drawbacks of this method a new rapid measurement technique was established which is the redox potential measurement method (Reichart et al., 2007).

THE REDOX POTENTIAL

The redox potential (Eh) is a physicochemical parameter which measures the oxidizing or reducing power of a system. The more the system is reducing (Eh negative), the more it can donate electrons. Likewise, for an oxidizing system, the more the Eh is positive, the more it can capture electrons. When two redox systems are in competition, the electrons go from the

system whose E_h is the lower (in algebraic value) to the one with the highest E_h (Jacob 1970).

The standard potential of a couple informs about their ability to give up or capture electrons which results in an electric change defined by Nernst's law which establishes the relation between the redox potential (E_h), and the oxidation and reduction activities of the chemical species involved in the reaction (Reichart et al. 2007)

$$E_h = E_o + \frac{RT}{nF} \cdot \ln \frac{(Ox.)}{(Red.)}$$

Where E_o is the half redox potential of the oxidizing / reduction couple for standard conditions (pressure = 1 atm, $T = 298$ K, pH 0, $[Ox] = [Red]$), R is the ideal gas constant, T is the absolute temperature, n refers to the number of electrons transferred, F is the Faraday constant, Ox and Red are the oxidized form activity of the species and the reduced form activity of the species respectively (Reichart et al. 2007).

REDOX POTENTIAL MEASUREMENT SYSTEM

The measuring electrode must be inert and acts as an acceptor or donor of electrons. The metal of the electrode can be made of gold or platinum having very high standard potentials, which allows to be electrochemically inert towards the medium (Vanýsek, 1998)

Platinum electrode presents a poor measurement reproducibility due to the presence of mineral and organic impurities, the formation of hydrogen peroxide and the oxidation of the platinum electrode on the surface. In this way, having a smooth platinum surface could improve the response time, the repeatability and the reproducibility of measurements (Jacob, 1970)

The reference electrodes most frequently used are the hydrogen electrode, the calomel electrode ($HgCl_2$) or the silver / chloride electrode. The two last electrodes are characterized by a constant potential due to the use of salts of low solubility so that the electrodes cannot be polarized (Jacob, 1970).

THE REDOX POTENTIAL MEASUREMENT METHOD

This method is based on the oxidation-reduction activity of microorganisms. The energy source of the microbial development is the biological oxidation which is accompanied by a reduction of the environment. This is related to the oxygen depletion and the production of reducing compounds in the nutrient medium. The reduction is the origin of an electrical change which can be measured continuously. This method can be carried out by multi-channel redox potential measuring instrument MicroTester where the electrodes are connected by a data collector (Szakmár et al., 2014).

The data collector automatically determines the TTD values from the redox curves and the first differential of the redox potential ($\Delta E/\Delta t$), TTD is the time required to reach the detection threshold when the value of the redox potential decrease and exceed the detection criterion (DC) (Erdosi et al., 2012). A strict linear correlation could be established between

the TTD and the logarithm of the viable count (log N) and/or metabolic rate of microorganisms (Engelhardt et al., 2018).

CONCLUSION

Redox potential measurement has been grabbing a lot of attention for quantitative and qualitative detection of bacteria. This method could be a great alternative to the conventional method since it is more simple and more rapid than the colony counting method.

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REFERENCES

- Engelhardt, T., Szakmár, K., Kiskó, G., Mohácsi-Farkas, C., & Reichart, O. (2018). Combined effect of NaCl and low temperature on antilisterial bacteriocin production of *Lactobacillus plantarum* ST202Ch. *LWT - Food Science and Technology*, 89, 104–109. <https://doi.org/10.1016/j.lwt.2017.10.027>
- Erdosi, O., Szakmár, K., Reichart, O., Székely-Körmöczy, P., & Laczay, P. (2012). Application of the redox potential measurement based rapid method in the microbial hygienic control. *Acta Alimentaria*, 41(1), 45–55. <https://doi.org/10.1556/AAlim.2011.0005>
- Feng, P. (1996). Emergence of Rapid Methods for Identifying Microbial Pathogens in Foods. *Journal of AOAC International*, 79(3), 809–812. <https://doi.org/10.1093/jaoac/79.3.809>
- Jacob, H. E. (1970). Redox Potential. *Methods in Microbiology*. [https://doi.org/10.1016/S0580-9517\(08\)70218-6](https://doi.org/10.1016/S0580-9517(08)70218-6)
- Jasson, V., Jacxsens, L., Luning, P., Rajkovic, A., & Uyttendaele, M. (2010). Alternative microbial methods: An overview and selection criteria. *Food Microbiology*, 27(6), 710–730. <https://doi.org/10.1016/j.fm.2010.04.008>
- Reichart, O., Szakmár, K., Jozwiak, Á., Felföldi, J., & Baranyai, L. (2007). Redox potential measurement as a rapid method for microbiological testing and its validation for coliform determination. *International Journal of Food Microbiology*, 114(2), 143–148. <https://doi.org/10.1016/j.ijfoodmicro.2006.08.016>
- Szakmár, K., Reichart, O., Szatmári, I., Erdosi, O., Szili, Z., László, N., Székely Körmöczy, P., & Laczay, P. (2014). In vitro study on the effect of doxycycline on the microbial activity of soil determined by redox-potential measuring system. *Acta Microbiologica et Immunologica Hungarica*, 61(3), 317–328. <https://doi.org/10.1556/AMicr.61.2014.3.6>
- Vanýsek, P. (1998). Equivalent Conductivity of Electrolytes in Aqueous Solution. In *CRC Handbook of Chemistry and Physics*, 79th edn.

THE EFFECT OF MILD HEAT TREATMENT ON SARCOPLASMIC AND MYOFIBRILLAR PROTEIN SOLUBILITY OF MEAT

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Summary: *Minimal processing technologies such as mild heat treatment present an emerging technology to produce high-protein foods such as chicken breast. The advantage of temperature and time selection during processing steps has a great effect on quality and functional attributes of the chicken breast such as moisture content, weight loss, and protein solubility. In this study, double-step sous vide treatments provided better quality attributes (moisture content, weight loss, cooking yield) compared with the single-step sous vide treatment. On the other hand, protein solubility was increased in double-step sous vide treated samples compared to the single-step sous vide treated chicken breast. Based on our results, double-step sous vide presented a valuable technology to enhance the quality and functional attributes of cooked chicken breast.*

Keywords: *Sous vide, sarcoplasmic protein solubility, cooking yield, moisture content.*

INTRODUCTION

Nowadays minimal processing techniques are having a steadily increasing application in the ready-to-eat food industry, catering sector, and households. Sous vide technology represents a mild thermal treatment of food performed at a specified time and temperature conditions under anaerobic conditions (Baldwin, 2012). Compared to the traditional techniques such as boiling, grilling, etc., sous vide treated foods (e.g. meat) provide higher moisture content, improved water-holding capacity, lower cooking loss, and better texture properties (Gluchowski et.al., 2020). During the heating process, meat proteins get denatured which leads to different changes in protein-water interaction, shrinking of muscle fibers (Zielbauer et.al., 2016). The application of high cooking temperatures leads to greater changes in protein structure causing transverse and longitudinal shrinking which has a direct effect on the water content of meat. These changes in protein structure are responsible for changes in quality attributes such as water-holding capacity, color, and texture attributes in meat (Lawrie & Ledward, 2014; Baily & Light, 1989). In this sense, careful selection of temperature and time in sous vide treatments would provide a valuable option to control the denaturation of different proteins responsible for different quality properties of the meat. These quality attributes of meat are linked with the functional properties of meat such as protein solubility. Murphy & Marks (2000) reported a direct relation between protein solubility and cooking loss of heat-treated chicken breast patties. The objective of the current study was to examine

the effect of different combinations of temperatures and times on the sarcoplasmic and myofibrillar protein solubility of mild heat-treated chicken breast and its relationship with moisture content and weight loss.

MATERIAL AND METHODS

Experimental design and treatment procedure

Chicken breast muscles (*pectoralis major*) purchased from a local market in Budapest (Hungary) were skinned off from fats and connective tissues. The samples were cut into uniform weight (130 ± 4 g) and thickness (2 ± 0.2 cm) and were grouped into 5 different groups. All the group samples were vacuum packaged and then subjected to heat treatment using a water bath. T1 group samples were treated at a single-step temperature of 60 °C, T2 and T3 at the first temperature of 45 °C and after at 60°C, T4 and T5 at the first temperature of 50°C and after at 60°C.

Table 1. Mild heat treatments steps applied in the study

Treatment groups	Time at the first step temperature of 45°C (min)	Time at the first step temperature of 50°C (min)	Time at the end step temperature of 60 °C (min)
T1	-	-	120
T2	40	-	80
T3	60	-	60
T4	-	40	80
T5	-	60	60

After treatment, the samples were rapidly cooled in ice water for 1h at (1°C) and held in refrigerated conditions (4°C) till the following day before analysis. Moisture content, cooking loss, and protein solubility were determined. All measurements were performed in triplicates.

Moisture content, cooking yield, and weight loss

Moisture content (%) was determined by calculating the difference of the weight of the samples before and after oven drying at 105 °C for 16 h (AOAC, 2005). Cooking yield and weight loss were calculated by subtracting the sample weight before and after sous vide treatments.

Protein solubility

Protein solubility of treated chicken breast samples was analyzed following the method as described by Warner et.al., 1997. For sarcoplasmic protein solubility determination, 1 g of chicken breast sample was mixed with 10 ml 0.025 M potassium phosphate buffer (pH=7.2). The mixtures were homogenized using Digital Ultra-Turrax Homogeniser (Staufen, Germany) and then were put at 4°C for 20 h. After that, the protein solubility was assessed using bovine serum albumin standard (Bradford, 1976). Total protein solubility followed the same procedure by using 0.55 M potassium iodide, 0.05 M potassium phosphate buffer (pH=7.2). Myofibrillar protein solubility was calculated from the difference of total protein solubility and sarcoplasmic protein solubility of the samples.

Statistical analysis

Statistical analysis was performed by SPSS-23 software (Version 23.0, SPSS INC., Chicago, IL, USA). Significant differences between the groups were analyzed through analysis of variance (ANOVA) using the Tukey post hoc test.

RESULTS AND DISCUSSION

Moisture content, cooking yield, and weight loss of chicken breast treated at a different time and temperature combinations are presented in Figure 1. Double-step sous vide treated chicken breast (T2-T5) showed higher moisture content and cooking yield as compared with the single-step sous vide (T1). This might be due to the lower first step temperature used in the double-step sous vide treatments. On the other hand, weight loss of double-step sous vide treated chicken breast was significantly lower compared with the single-step sous vide ($P \leq 0.05$).

Significant differences in cooking yield and weight loss were also observed between double-step sous vide treatments where the first step temperature was 45°C (T2-T3) and double-step sous vide treatments where the first-step temperature was 50°C (T4-T5). Both cooking yield and weight loss are related with juiciness of the meat which present a desired sensory attribute by consumers. In this sense, double-step sous vide treated chicken breasts are likely to be more attractive for consumers compared to single-step sous vide treated chicken breast.

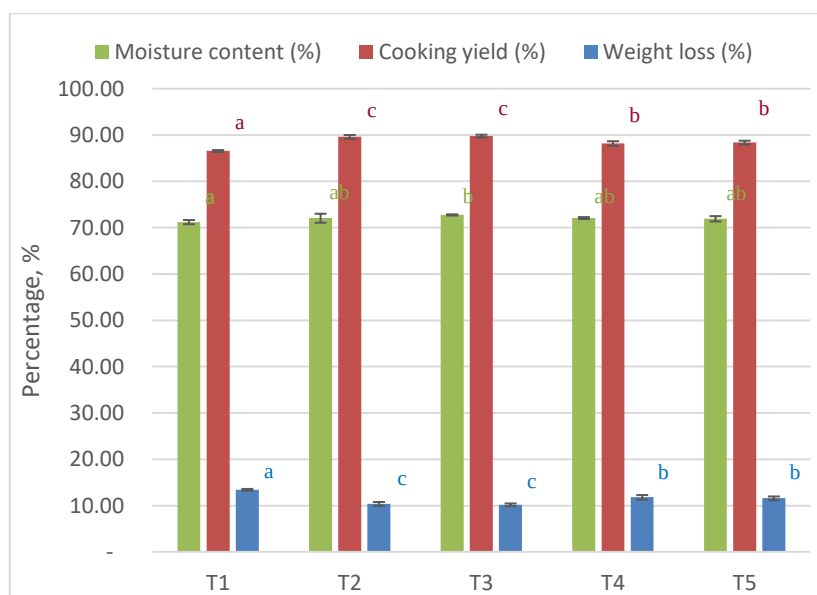


Figure 1. Average values of moisture content (%) and cooking yield (%) and weight loss (%) of the mild heat-treated chicken breast under different time and temperature combinations

T1 - 60° C for 120 min; T2 - 45° C for 40 min and 60° C for 80 min; T3 - 45° C for 60 min and 60° C for 60 min; T4 - 50° C for 40 min and 60° C for 80 min; T5 - 50° C for 60 min and 60° C for 60 min; Different letters above the bars indicate a significant difference ($P \leq 0.05$).

According to the results, total protein solubility was significantly increased in double-step sous vide treatments (T2-T5) as compared to the single-step sous vide treatment (T1). The use of low temperatures as first step temperature in double-step sous vide treatments showed higher protein solubility, moisture content and the lowest weight loss. No significant differences were observed in myofibrillar protein solubility between single-step and double-step sous vide treatments. With the exception of T2 treatment, sarcoplasmic protein solubility was significantly increased in case of double-step treatments compared to single-step mild heat treatments.

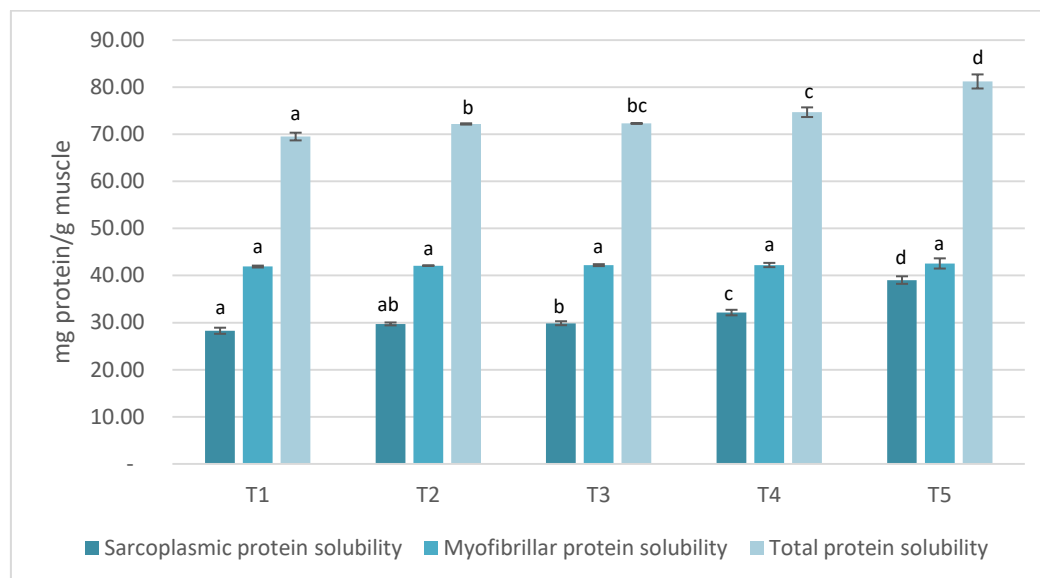


Figure 2. Sarcoplasmic protein solubility, myofibrillar protein solubility, and total protein solubility in mild heat treated chicken breast under different time and temperature combinations

T1 - 60° C for 120 min; T2 - 45° C for 40 min and 60° C for 80 min; T3 - 45° C for 60 min and 60° C for 60 min; T4 - 50° C for 40 min and 60° C for 80 min; T5 - 50° C for 60 min and 60° C for 60 min; Different letters above the bars indicate a significant difference ($P \leq 0.05$). There was a correlation between total protein solubility and weight loss in mild heat-treated chicken breast (data not shown). The correlation of weight loss with sarcoplasmic protein solubility was higher compared to myofibrillar protein solubility, which agrees with the observation of Joo et.al., (1999). The relationship between protein solubility and water content is often used to provide more information about protein denaturation process in cooked meat (Mudalal et.al., 2014).

CONCLUSION

In our study, we examined the effect the mild heat treatment using a low first step temperature on quality and functional properties of cooked chicken breast. We found that double-step mild heat treatments provided better quality properties such as higher moisture content, cooking yield and weight loss as compared with the single-step treatment. Moreover, chicken breast cooked with double-step mild heat treatments showed higher sarcoplasmic and

total protein solubility than single-step mild heat-treated chicken breast. In addition to the studied quality properties, it is necessary to examine the changes on color, texture, oxidative stability, protein denaturation and conduct microbial analysis to determine the safety of the selected mild heat treatments.

REFERENCES

- BAILEY, A.J. and LIGHT, N.D., (1989). *Connective tissue in meat and meat products*. Elsevier applied science.
- BALDWIN, D.E., (2012). Sous vide cooking: A review. *International Journal of Gastronomy and Food Science*, 1(1), pp.15-30.
- GLUCHOWSKI, A., CZARNIECKA-SKUBINA, E. and BULA, M., (2020). The use of the sous-vide method in the preparation of poultry at home and in catering—Protection of nutrition value whether high energy consumption. *Sustainability*, 12(18), p.7606.
- JOO, S.T., KAUFFMAN, R.G., KIM, B.C. and PARK, G.B., (1999). The relationship of sarcoplasmic and myofibrillar protein solubility to colour and water-holding capacity in porcine longissimus muscle. *Meat science*, 52(3), pp.291-297.
- LAWRIE, R.A. and LEDWARD, D., (2014). *Lawrie's meat science*. Woodhead Publishing.
- MUDALAL, S., BABINI, E., CAVANI, C. and PETRACCI, M., (2014). Quantity and functionality of protein fractions in chicken breast fillets affected by white striping. *Poultry science*, 93(8), pp.2108-2116.
- MURPHY, R.Y. and MARKS, B.P., (2000). Effect of meat temperature on proteins, texture, and cook loss for ground chicken breast patties. *Poultry science*, 79(1), pp.99-104.
- ZIELBAUER, B.I., FRANZ, J., VIEZENS, B. and VILGIS, T.A., (2016). Physical aspects of meat cooking: time dependent thermal protein denaturation and water loss. *Food biophysics*, 11(1), pp.34-42.

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